

THE GENERALIZED GAUSSIAN ITERATED MAP*

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Abstract. We develop an analytical study of a generalized Gaussian iterated map, a four-parameter family of smooth nonlinear maps that reduces, through an affine conjugacy, to a normalized two-parameter family. The normalized map has a unique nondegenerate quadratic critical point and negative Schwarzian derivative away from it. We derive explicit parameterizations of the fixed-point multiplier $+1$ and -1 loci, establish their saddle-node and period-doubling nondegeneracy properties, and identify a codimension-two cusp on the saddle-node locus. Both loci attain the common minimum, which coincides with the global contraction threshold. We obtain a complete classification of the number of fixed points and, using Singer's theorem, prove that when the critical point lies in the interior of the invariant interval the map admits at most two attracting periodic orbits; an attracting orbit whose immediate basin avoids the endpoints is unique among such orbits and necessarily attracts the critical point. We also construct an explicit one-parameter family satisfying the Collet–Eckmann condition, yielding an ergodic absolutely continuous invariant probability measure with a physical basin of positive Lebesgue measure. Finally, we derive the Perron–Frobenius operator in closed form and, under explicit regularity and noncollision assumptions, characterize one-sided square-root singularities of invariant densities, their propagation along the post-critical orbit, their amplitudes, and a finite local post-critical decomposition.

Key words. nonlinear iterated maps, Gaussian nonlinearity, unimodal dynamics, chaos, bifurcation analysis, interval maps

AMS subject classifications. 37E05, 37C05, 37C25, 37G15

1. Introduction. Discrete dynamical systems generated by iterating nonlinear maps play a central role in the mathematical description of nonlinear phenomena [1, 16, 37]. They arise naturally in sampled-data systems [40], nonlinear feedback mechanisms [29], population dynamics [24], signal processing [31], and control applications [25]. Despite their apparent simplicity, iterated maps exhibit an extraordinary range of behaviors, including bifurcations, periodic oscillations, strange attractors, and deterministic chaos. As a result, they have become a fundamental paradigm for studying nonlinear dynamics. Classical examples include the logistic map [24], the Hénon map [17, 5, 6], and the Chirikov standard map [11].

Among smooth unimodal maps, the logistic family has become the canonical model for investigating bifurcation theory, universality, invariant measures, and statistical properties [9, 37]. Nevertheless, many nonlinear phenomena encountered in physics and engineering are governed by localized nonlinear responses rather than polynomial nonlinearities. Gaussian nonlinearities naturally arise in kernel methods, nonlinear filtering, optical systems, neural computation, and localized amplification mechanisms [8, 31]. These considerations motivate the study of Gaussian iterated maps as a natural counterpart of polynomial unimodal families.

In this work we consider the generalized Gaussian map

$$(1.1) \quad x_{n+1} = f(x_n), \quad f(x) = \gamma e^{-\alpha(x-\delta)^2} + \beta,$$

where $\alpha > 0$ and $\beta, \gamma, \delta \in \mathbb{R}$. This four-parameter family generalizes the classical

*A preliminary numerical study of the generalized Gaussian map appeared in [38]. The present work substantially extends that study through a rigorous analytical treatment of its dynamic behavior and mathematical structure.

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Gaussian map

$$(1.2) \quad x_{n+1} = e^{-\alpha x_n^2} + \beta, \quad \alpha > 0,$$

obtained for $\gamma = 1$ and $\delta = 0$. The generalized formulation was introduced in [38] and allows the nonlinear response geometry and location to be controlled independently.

Each parameter in (1.1) has a distinct geometric and dynamical role. The parameter α controls the width of the Gaussian profile: increasing α produces a narrower and more localized nonlinear response, whereas decreasing α broadens it. The parameter γ determines the signed amplitude of the Gaussian term, controlling the strength of the nonlinear feedback. For $\gamma > 0$, the map has a single maximum at $x = \delta$, while for $\gamma < 0$ the profile is inverted and $x = \delta$ becomes a minimum. The parameter δ shifts the Gaussian horizontally, fixing the location of the unique critical point and therefore the region where the nonlinear folding occurs. Finally, β introduces a vertical offset, determining the asymptotic level of the map,

$$\lim_{|x| \rightarrow \infty} f(x) = \beta,$$

and shifting the image interval without changing its shape. Together, the four parameters provide independent control of the width, amplitude, position, and offset of the nonlinear response, yielding a flexible family of maps with a rich dynamical behavior.

Through an affine change of variables, the four-parameter system (1.1) can be reduced to the normalized family

$$y_{n+1} = A + Be^{-y_n^2},$$

which depends only on two effective parameters while preserving the qualitative dynamics of the original system. At the same time, it exhibits several analytical features that distinguish it from polynomial families. In particular, its inverse branches admit explicit closed-form expressions, allowing the Perron-Frobenius operator to be written explicitly [2]. The Gaussian family therefore combines the principal dynamical mechanisms of smooth unimodal maps with an analytical structure that permits many of these mechanisms to be described in closed form. This analytical tractability provides an unusual opportunity to investigate, within a single framework, the interplay between local geometry, global dynamics, and statistical properties.

Although Gaussian maps have been employed in several applications, including critical slowing down and bifurcation analysis [15], image processing and information security [33, 34], and generalized Gaussian models arising in nonextensive statistical mechanics [7, 10, 30, 35], their mathematical theory remains considerably less developed than that of classical polynomial families. Existing studies are predominantly numerical and focus mainly on bifurcation diagrams and chaotic behavior, whereas a systematic analytical treatment of the parameter space, invariant measures, transfer operators, and statistical properties is still lacking.

The Gaussian map differs globally from polynomial unimodal maps. Whereas the logistic map is unbounded outside a restricted phase-space interval, the Gaussian nonlinearity decays exponentially and maps the real line into a compact interval determined by the parameters. Thus, the Gaussian family combines a local quadratic fold with a bounded global image and exponentially decaying derivatives in the tails.

Relation with the classical theory of interval maps. The qualitative theory of smooth unimodal interval maps is by now well developed. The monograph of

de Melo and van Strien provides a systematic framework for the study of nonflat critical points, negative Schwarzian derivative, kneading theory, bifurcations, and rigidity phenomena [14]. Within this framework, Singer’s theorem shows that, for a sufficiently smooth interval map with negative Schwarzian derivative, the immediate basin of every attracting periodic orbit contains either a critical point or a boundary point [36]. These results are general structural theorems: they apply to broad classes of interval maps but do not usually provide explicit parameter boundaries or closed formulas for a particular nonlinear family. In the present work, the Gaussian structure makes it possible to convert this general theory into explicit statements, including analytical saddle-node and period-doubling curves, a complete fixed-point count, and an upper bound on the number of attracting periodic orbits together with a conditional uniqueness criterion.

The existence of chaotic parameter regimes with absolutely continuous invariant measures is also a classical theme in one-dimensional dynamics. Jakobson proved that positive-measure sets of parameters in the quadratic family admit absolutely continuous invariant probability measures [19]. Collet and Eckmann introduced a quantitative nonuniform-expansion condition based on exponential derivative growth along the critical orbit [12], and subsequent work of Nowicki and van Strien established the existence of absolutely continuous invariant measures for broad classes of unimodal maps satisfying this condition [26]. Keller and Nowicki further developed the spectral theory of transfer operators for Collet-Eckmann maps [22]. These results establish statistical behavior under abstract dynamical hypotheses. Our contribution is different in emphasis: we construct explicit parameter curves for which the Gaussian critical orbit satisfies the relevant expansion mechanism and then exploit the closed form of the map to analyze the associated density singularities directly.

The operator-theoretic study of invariant densities originates in the classical work of Lasota and Yorke on piecewise monotone expanding transformations [23], and was developed further in the theory of piecewise expanding and piecewise continuous interval maps [2, 9, 20, 32]. In that setting, Lasota-Yorke inequalities, bounded-variation estimates, and spectral arguments provide existence, stability, and convergence results for invariant densities. Smooth unimodal maps with critical points require additional care, because the derivative vanishes and the invariant density generally develops singularities along the post-critical orbit. The present Gaussian family is particularly tractable in this respect: its two inverse branches are explicit, and therefore the Perron-Frobenius operator, the invariant-density equation, and the singularity propagation law can all be written in closed form.

Modern transfer-operator theory provides general results on spectral and statistical stability, decay of correlations, and linear response [2, 3, 21, 39]. The present work does not reproduce this general theory; instead, it exploits the closed-form Gaussian structure to describe directly the two-preimage contribution, the Jacobian degeneracy at the critical value, and the propagation of post-critical density singularities.

Objectives, contributions and paper organization. The purpose of the present work is to develop a systematic analytical framework for the generalized Gaussian map. Our approach combines techniques from one-dimensional dynamics, bifurcation theory, and Perron-Frobenius operator theory to establish both deterministic and statistical properties of the system. Particular emphasis is placed on exploiting the explicit analytical structure of the Gaussian map to derive results that are usually inaccessible for general smooth unimodal maps.

The main contributions of this work are summarized as follows:

1. We reduce the generalized four-parameter Gaussian map to a normalized two-parameter family and establish its fundamental geometric properties, including the existence of a unique nondegenerate quadratic critical point and a negative Schwarzian derivative away from the critical point;
2. We derive explicit analytical expressions for the saddle-node and fixed-point period-doubling bifurcation curves, obtaining a complete analytical characterization of the parameter regions containing one or three equilibria;
3. We establish rigorous global dynamical properties of the normalized Gaussian map, including a bound of at most two attracting periodic orbits when the critical point lies in the interior of the invariant interval, a uniqueness criterion for attracting orbits whose immediate basins avoid the endpoints, and explicit parameter curves satisfying the Collet–Eckmann condition;
4. Using the explicit inverse branches of the Gaussian map, we derive the Perron-Frobenius operator in closed form together with the corresponding invariant-density equation;
5. Under explicit regularity and noncollision assumptions, we obtain a detailed analytical description of invariant-density singularities, derive their propagation along finite post-critical segments and their amplitudes, and establish a finite local decomposition near the corresponding post-critical points;
6. The theoretical developments are complemented throughout the paper by high-resolution numerical experiments illustrating the geometry of the map, the analytical parameter portrait, bifurcation structure, invariant densities, Perron-Frobenius evolution, and singularity propagation.

The remainder of the paper is organized as follows. Section 2 introduces the normalized Gaussian map and develops its local geometric properties. Section 3 establishes the analytical bifurcation structure and parameter-plane geometry. Section 4 investigates the global dynamics and the onset of nonuniformly expanding behavior. Section 5 develops the Perron-Frobenius theory, derives the invariant-density equation, and analyzes the propagation of singularities. Section ?? discusses the significance, limitations, and broader implications of the results. Finally, Section 7 summarizes the main conclusions and outlines directions for future work.

2. Basic analytical properties of the map. Before analyzing the dynamical regimes of (1.1), it is important to establish some structural analytical properties of the map. These properties allow us to place the system within the classical framework of one-dimensional nonlinear dynamics and provide the mathematical foundation for the bifurcation analysis developed in later sections.

In particular, we show that the map is smooth and bounded, that its dynamics are confined to a compact invariant interval, and that it possesses a unimodal structure with a single critical point. We also identify a parameter regime in which the system contracts and therefore admits a unique globally attracting fixed point. Finally, we show that the generalized Gaussian map can be reduced, via an affine transformation, to a normalized two-parameter family that retains the essential dynamical features of the original system.

2.1. Smoothness and invariant interval. We begin by examining the map’s basic regularity and range. These properties are fundamental because they ensure that the iterates of the map remain confined to a bounded region of phase space, allowing the dynamics to be studied using the well-developed theory of interval maps [13, 14].

THEOREM 2.1 (Smoothness and bounded range of the map). *The map $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by*

$$(2.1) \quad f(x) = \gamma e^{-\alpha(x-\delta)^2} + \beta, \quad \alpha > 0,$$

is C^∞ and bounded. Moreover,

$$\lim_{|x| \rightarrow \infty} f(x) = \beta,$$

and

$$f(\mathbb{R}) = \begin{cases} (\beta, \beta + \gamma], & \gamma > 0, \\ \{\beta\}, & \gamma = 0, \\ [\beta + \gamma, \beta), & \gamma < 0. \end{cases}$$

Proof. Since the exponential function is C^∞ , the map f is also C^∞ . Because $\alpha > 0$, we have $e^{-\alpha(x-\delta)^2} \in (0, 1]$ for all $x \in \mathbb{R}$, with the value 1 attained uniquely at $x = \delta$ and with $e^{-\alpha(x-\delta)^2} \rightarrow 0$ as $|x| \rightarrow \infty$. Consequently, $\gamma e^{-\alpha(x-\delta)^2}$ takes all values in $(0, \gamma]$ when $\gamma > 0$, equals 0 identically when $\gamma = 0$, and takes all values in $[\gamma, 0)$ when $\gamma < 0$. Adding the constant β shifts these ranges to the stated values.

Furthermore, since $e^{-\alpha(x-\delta)^2} \rightarrow 0$, we have

$$\lim_{|x| \rightarrow \infty} f(x) = \beta. \quad \square$$

This theorem shows that the image of the real line under f is a bounded interval. This observation immediately leads to an invariant region that contains all forward trajectories.

COROLLARY 2.2 (Existence of an invariant interval). *Let*

$$I = \begin{cases} [\beta, \beta + \gamma], & \gamma > 0, \\ \{\beta\}, & \gamma = 0, \\ [\beta + \gamma, \beta], & \gamma < 0. \end{cases}$$

Then $f(\mathbb{R}) \subset I$, $f(I) \subset I$, and every orbit enters I after at most one iteration.

Proof. We treat the case $\gamma > 0$; the case $\gamma < 0$ is analogous, and $\gamma = 0$ is trivial.

Step 1: $f(\mathbb{R}) \subset I$. From Theorem 2.1, $f(\mathbb{R}) = (\beta, \beta + \gamma] \subset [\beta, \beta + \gamma] = I$.

Step 2: $f(I) \subset I$. Since $I \subset \mathbb{R}$, we have

$$f(I) \subset f(\mathbb{R}) \subset I,$$

where the second inclusion follows from Step 1.

Step 3: *Every orbit enters I in one step.* For any initial condition $x_0 \in \mathbb{R}$, Step 1 gives $x_1 = f(x_0) \in I$. By Step 2, all subsequent iterates $x_n = f(x_{n-1})$ remain in I for all $n \geq 1$. $\square$

This result shows that the long-term behavior of the system can be studied as a dynamical system on a compact interval. Such a reduction is important because many classical results in nonlinear dynamics are formulated for continuous maps on compact subsets of the real line.

2.2. Critical point and unimodality. The qualitative structure of a one-dimensional map is strongly influenced by its critical points. For the generalized Gaussian map, the shape of the exponential nonlinearity suggests the existence of a single peak. The following result confirms that the map has exactly one critical point.

PROPOSITION 2.3. *The derivative of f in (2.1) is*

$$f'(x) = -2\alpha\gamma(x - \delta)e^{-\alpha(x-\delta)^2}.$$

If $\gamma \neq 0$, then $f'(x) = 0$ if and only if $x = \delta$, so $x = \delta$ is the unique critical point. If $\gamma = 0$, the map is constant and $f'(x) = 0$ for every $x \in \mathbb{R}$.

Proof. Differentiating f gives

$$f'(x) = \gamma(-2\alpha(x - \delta))e^{-\alpha(x-\delta)^2}.$$

If $\gamma \neq 0$, then, since the exponential term never vanishes, $f'(x) = 0$ if and only if $x = \delta$. If $\gamma = 0$, then $f' \equiv 0$. $\square$

The existence of a single critical point implies that the map belongs to the class of unimodal maps whenever $\gamma > 0$. Unimodal maps play a central role in the theory of nonlinear dynamics because they exhibit many of the standard routes to chaos.

PROPOSITION 2.4. *If $\gamma > 0$, the map f in (2.1) is strictly increasing on $(-\infty, \delta)$ and strictly decreasing on (δ, ∞) . Hence f is unimodal with a global maximum at $x = \delta$.*

Proof. From the expression for f' , we see that the sign of the derivative is the sign of $-(x - \delta)$. Therefore $f'(x) > 0$ when $x < \delta$ and $f'(x) < 0$ when $x > \delta$. Thus, f increases to $x = \delta$ and then decreases, giving a single maximum. $\square$

The unimodal structure identified above places the Gaussian map within the class of one-dimensional nonlinear systems whose dynamics are well known to include bifurcations, periodic windows, and chaotic regimes [13, 14].

2.3. Fixed points and contraction regime. We now turn to the existence and stability of fixed points. Fixed points represent equilibrium states of the discrete dynamical system and often provide the simplest attractors of the dynamics.

PROPOSITION 2.5. *The map f in (2.1) always has at least one fixed point.*

Proof. Define $g(x) = f(x) - x$. Then

$$\lim_{x \rightarrow -\infty} g(x) = +\infty, \quad \lim_{x \rightarrow +\infty} g(x) = -\infty.$$

Since g is continuous, the intermediate value theorem implies that g has at least one zero. Hence f has at least one fixed point. $\square$

The stability of these fixed points depends on the magnitude of the derivative. The next result provides a global bound on the map's derivative.

PROPOSITION 2.6. *The derivative of f in (2.1) satisfies the bound*

$$|f'(x)| \leq |\gamma|\sqrt{2\alpha}e^{-1/2}.$$

Proof. Let $u = x - \delta$. Then

$$|f'(x)| = 2\alpha|\gamma||u|e^{-\alpha u^2}.$$

The function $|u|e^{-\alpha u^2}$ attains its maximum at $|u| = 1/\sqrt{2\alpha}$, giving

$$\max |f'(x)| = |\gamma|\sqrt{2\alpha}e^{-1/2}.$$

□

The bound above allows us to identify a parameter region where the map becomes a contraction.

THEOREM 2.7 (Global contraction criterion). *If*

$$|\gamma|\sqrt{2\alpha}e^{-1/2} < 1,$$

then f in (2.1) is a global contraction. Consequently, the fixed point is unique and globally attracting.

Proof. Proposition 2.6 shows that $|\gamma|\sqrt{2\alpha}e^{-1/2}$ is a global Lipschitz constant of f . If this constant is strictly less than one, as stated in the hypothesis, the result therefore follows from Banach Fixed point Theorem (see Theorem A.6 in the appendix). □

Theorem 2.7 identifies a region of parameter space

$$\mathcal{R} = \left\{ (\alpha, \beta, \gamma, \delta) \in \mathbb{R}^4 : \alpha > 0, |\gamma|\sqrt{\alpha} < \sqrt{\frac{e}{2}} \right\},$$

in which the dynamics of the generalized Gaussian map system are simple and predictable. In this regime, all trajectories converge to a single equilibrium, and more complex dynamical phenomena such as oscillations or chaos cannot occur.

2.4. Affine normalization. Although the map (1.1) contains four parameters, its essential dynamical structure can be simplified by an appropriate change of variables. Such a reduction allows the map family to be studied in a lower-dimensional parameter space.

THEOREM 2.8. *The generalized Gaussian map (2.1) is affinely conjugate to the normalized Gaussian family*

$$(2.2) \quad T(y) = A + Be^{-y^2},$$

via the affine diffeomorphism $\phi : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\phi(x) = \sqrt{\alpha}(x - \delta),$$

with parameters

$$A = \sqrt{\alpha}(\beta - \delta), \quad B = \sqrt{\alpha}\gamma.$$

That is, $\phi \circ f = T \circ \phi$, or equivalently the following diagram commutes:

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{f} & \mathbb{R} \\ \phi \downarrow & & \downarrow \phi \\ \mathbb{R} & \xrightarrow{T} & \mathbb{R} \end{array}$$

In particular, f and T are topologically conjugate, and their dynamics are fully equivalent: ϕ maps orbits of f to orbits of T , preserving the period and stability type of every periodic orbit.

Proof. We verify the conjugacy relation $\phi \circ f = T \circ \phi$ directly.

Step 1: ϕ is an affine homeomorphism. The map $\phi(x) = \sqrt{\alpha}(x - \delta)$ is affine with slope $\sqrt{\alpha} > 0$ (since $\alpha > 0$), hence a homeomorphism of $\mathbb{R}$ onto itself. Its inverse is

$$\phi^{-1}(y) = \delta + \frac{y}{\sqrt{\alpha}}.$$

Step 2: Computing $(\phi \circ f)(x)$. Applying f first and then ϕ gives

$$(\phi \circ f)(x) = \phi\left(\gamma e^{-\alpha(x-\delta)^2} + \beta\right) = \sqrt{\alpha}\gamma e^{-\alpha(x-\delta)^2} + \sqrt{\alpha}(\beta - \delta).$$

Step 3: Computing $(T \circ \phi)(x)$. Setting $y = \phi(x) = \sqrt{\alpha}(x - \delta)$, so that $\alpha(x - \delta)^2 = y^2$, and applying T gives

$$(T \circ \phi)(x) = T(\sqrt{\alpha}(x - \delta)) = A + B e^{-(\sqrt{\alpha}(x-\delta))^2} = \sqrt{\alpha}(\beta - \delta) + \sqrt{\alpha}\gamma e^{-\alpha(x-\delta)^2}.$$

Step 4: Equality. Comparing Steps 2 and 3, we obtain

$$(\phi \circ f)(x) = (T \circ \phi)(x) \quad \text{for all } x \in \mathbb{R},$$

which establishes the conjugacy $\phi \circ f = T \circ \phi$, or equivalently $T = \phi \circ f \circ \phi^{-1}$.

Step 5: Dynamical consequences. Since the map ϕ is a homeomorphism satisfying $T = \phi \circ f \circ \phi^{-1}$, it follows by induction that

$$T^n = \phi \circ f^n \circ \phi^{-1} \quad \text{for all } n \geq 1,$$

where f^n and T^n denote the n -th iterates of f and T respectively. Consequently, x^* is a fixed point of f if and only if $\phi(x^*)$ is a fixed point of T , and more generally x^* is a periodic point of f of period n if and only if $\phi(x^*)$ is a periodic point of T of the same period n . Furthermore, since ϕ is differentiable with $\phi'(x) = \sqrt{\alpha} \neq 0$ everywhere, the chain rule gives

$$(f^n)'(x^*) = (T^n)'(\phi(x^*))$$

for every periodic point x^* of period n , so the multiplier of every periodic orbit is preserved under conjugacy, and stability types are identical for f and T . $\square$

This transformation shows that the four parameters of the original system reduce to two effective parameters A and B . The resulting normalized Gaussian family (2.2) retains the essential dynamical properties of the original map (2.1) while simplifying the analytical treatment.

Remark 2.9 (Reduction to $B > 0$). From Theorem 2.8, the parameter $B = \sqrt{\alpha}\gamma$ inherits the sign of γ , since $\sqrt{\alpha} > 0$. The case $B = 0$ (equivalently $\gamma = 0$) reduces f to the constant map $f(x) = \beta$, with trivial dynamics: every orbit reaches the fixed point β in one step. It therefore remains to consider $B \neq 0$.

We claim that the case $B < 0$ introduces no essentially new dynamics beyond the case $B > 0$. To see this, consider the affine involution $\psi : \mathbb{R} \rightarrow \mathbb{R}$ defined by $\psi(y) = -y$. If $T_{A,B}(y) = A + B e^{-y^2}$ with $B < 0$, then for every $y \in \mathbb{R}$,

$$\begin{aligned} (\psi \circ T_{A,B} \circ \psi^{-1})(y) &= -T_{A,B}(-y) \\ &= -\left(A + B e^{-(-y)^2}\right) \\ &= -A + (-B) e^{-y^2} \\ &= T_{-A, -B}(y). \end{aligned}$$

Since $B < 0$, we have $-B > 0$, so $T_{A,B}$ is affinely conjugate via ψ to $T_{-A,-B}$, which belongs to the family with positive second parameter. The conjugacy ψ is an affine homeomorphism satisfying $\psi^{-1} = \psi$, so by the same argument as in Theorem 2.8, it preserves the period and multiplier of every periodic orbit. In particular, the bifurcation structure of $T_{A,B}$ with $B < 0$ is identical to that of $T_{-A,-B}$ with $-B > 0$, with A replaced by $-A$.

Consequently, the full normalized parameter space $B \neq 0$ is covered by studying the $B > 0$ half-plane together with the substitution $(A, B) \mapsto (-A, -B)$. All subsequent results are therefore stated for $B > 0$ without loss of generality.

Before proceeding to the analytical study of the generalized Gaussian family bifurcation structure, it is useful to visualize how the normalized map changes as the parameters vary. Figure 1 presents four representative configurations illustrating the principal dynamical regimes that will be characterized rigorously in the following sections. These examples anticipate the global contraction regime, the emergence of multiple equilibria through saddle-node bifurcations, the local geometry at the period-doubling threshold, and a representative parameter set belonging to the explicit Collet-Eckmann family introduced later in Section 4.

Although the four panels illustrate qualitatively distinct dynamical regimes, they all belong to the same normalized family (2.2). The following sections establish rigorous conditions under which these different geometries arise, derive explicit bifurcation curves, bound the number of attracting periodic orbits, and characterize the onset of nonuniformly expanding dynamics.

2.5. Properties of the normalized map. We now examine some structural properties of the normalized Gaussian map (2.2). These properties play an important role in the subsequent dynamical analysis. In particular, we show that the map possesses a unique nondegenerate quadratic critical point and that its Schwarzian derivative is negative. Together, these features place the system within the classical class of smooth unimodal maps with negative Schwarzian derivative, for which several powerful results are available in one-dimensional dynamics.

PROPOSITION 2.10. *The normalized map T in (2.2) with $B \neq 0$ has a unique critical point at $y_{cr} = 0$, and*

$$T''(0) = -2B \neq 0.$$

Hence, this critical point is nondegenerate and quadratic.

Proof. Differentiating T yields

$$(2.3) \quad T'(y) = -2Bye^{-y^2}.$$

This derivative vanishes only when $y = 0$. Differentiating again gives

$$(2.4) \quad T''(y) = 2B(2y^2 - 1)e^{-y^2},$$

so

$$T''(0) = -2B \neq 0. \quad \square$$

The quadratic character of the critical point can be seen directly from the local Taylor expansion

$$T(y) = A + B - By^2 + \mathcal{O}(y^4), \quad y \rightarrow 0.$$

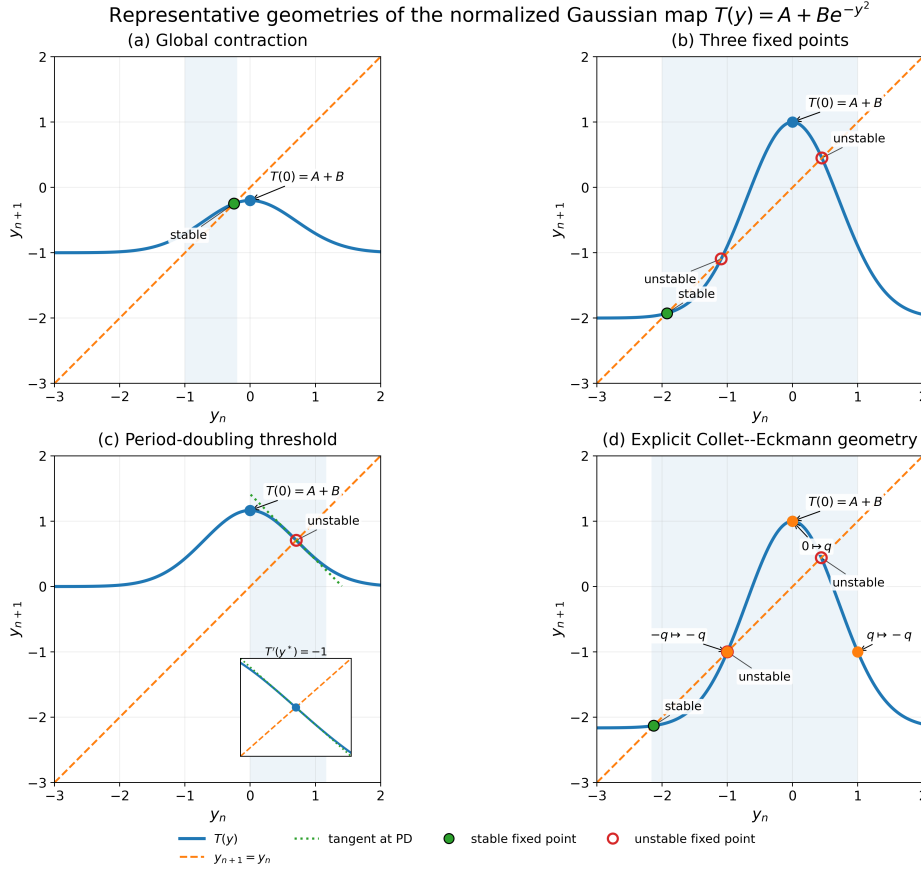


FIG. 1. Representative geometries of the normalized Gaussian map $T(y) = A + Be^{-y^2}$. Panel (a) shows a parameter set inside the global contraction region, where the map possesses a globally attracting fixed point. Panel (b) shows a parameter set with three coexisting fixed points; for the displayed parameter values, one is attracting and two are repelling. Panel (c) corresponds to the fixed-point period-doubling threshold, where the multiplier satisfies $\eta = -1$, corresponding to a generic fixed-point period-doubling (flip) bifurcation. Panel (d) presents a representative member of the explicit Collet-Eckmann family constructed in Section 4, for which the post-critical orbit $0 \mapsto q \mapsto -q \mapsto -q$ lands on a repelling fixed point. The shaded regions indicate the invariant interval $I = [A, A + B]$, the dashed line denotes the identity map, and the fixed-point stability is indicated at each equilibrium. Together these representative examples provide geometric intuition for the analytical results developed in Sections 3–5.

Consequently, the dynamics in a neighborhood of the critical point are governed, to leading order, by a quadratic fold, placing the normalized Gaussian map in the standard class of smooth unimodal maps with a quadratic critical point.

Figure 2 provides a geometric illustration of this local structure by comparing the exact Gaussian map with its quadratic Taylor approximation and by quantifying the corresponding approximation error. The agreement near $y = 0$ confirms geometrically that the leading nonlinear term is quadratic. This local normal form is the source of the universal square-root singularity derived later for the invariant density.

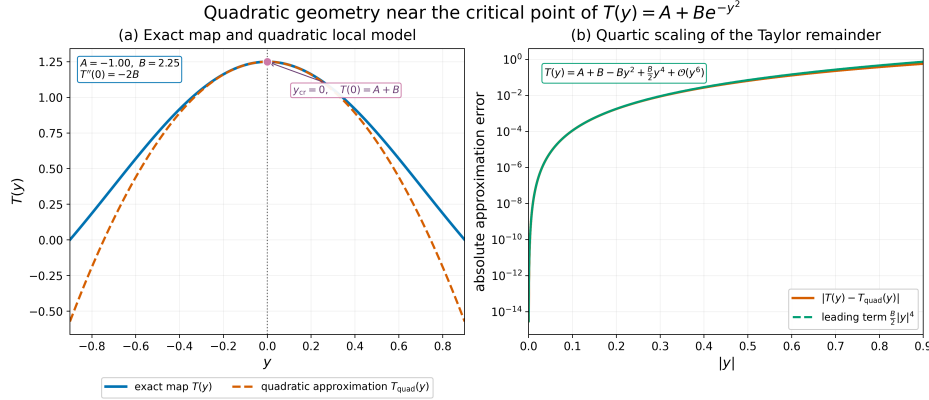


FIG. 2. Quadratic geometry near the critical point of the normalized Gaussian map $T(y) = A + Be^{-y^2}$. (a) Comparison between the exact Gaussian map and its quadratic Taylor approximation, showing that the two curves are locally indistinguishable in a neighborhood of the critical point $y_{cr} = 0$. The point $(0, A + B)$ is the unique critical point and corresponds to the maximum of the map. (b) Absolute approximation error together with the leading-order asymptotic term. The quartic scaling verifies that the quadratic approximation captures the dominant local geometry responsible for the folding mechanism of the Gaussian map.

A further important structural property of the normalized map concerns its Schwarzian derivative

$$(2.5) \quad S(T) = \frac{T'''}{T'} - \frac{3}{2} \left(\frac{T''}{T'} \right)^2.$$

The sign of the Schwarzian derivative plays a significant role in the theory of one-dimensional dynamics, as it imposes strong constraints on the number and types of attracting periodic orbits [14].

PROPOSITION 2.11. The Schwarzian derivative of T in (2.2) is given by

$$S(T)(y) = -2y^2 - \frac{3}{2y^2}, \quad y \neq 0.$$

In particular,

$$S(T)(y) < 0 \quad \text{for all } y \neq 0.$$

Proof. Since the first and second derivatives of T have already been computed, it remains only to evaluate the third derivative and substitute the resulting expressions into the definition of the Schwarzian derivative (2.5). A straightforward differentiation gives

$$(2.6) \quad T'''(y) = 4By(3 - 2y^2)e^{-y^2}.$$

Consequently, for every $y \neq 0$ one obtains

$$\frac{T'''(y)}{T'(y)} = 4y^2 - 6$$

and

$$\frac{T''(y)}{T'(y)} = \frac{1}{y} - 2y.$$

Substituting these identities into (2.5) yields

$$S(T)(y) = (4y^2 - 6) - \frac{3}{2} \left(\frac{1}{y} - 2y \right)^2.$$

After simplification, this becomes

$$S(T)(y) = -2y^2 - \frac{3}{2y^2}.$$

Since both terms on the right-hand side are strictly negative for $y \neq 0$, it follows immediately that

$$S(T)(y) < 0 \quad \text{for all } y \neq 0.$$

This completes the proof. $\square$

The negativity of the Schwarzian derivative is a fundamental structural property of the normalized Gaussian family (2.2). In the theory of one-dimensional dynamics, smooth unimodal maps with negative Schwarzian derivative form a particularly important class, since this condition imposes strong restrictions on the number and nature of attracting periodic orbits and plays a central role in the organization of bifurcation scenarios [36] (see Theorem A.7 in the appendix). Accordingly, Proposition 2.11 places the Gaussian family within the classical framework of smooth unimodal dynamics and justifies the use of standard analytical tools from that theory.

3. Equilibria and Bifurcation Analysis. In this section, we investigate the equilibrium structure of the generalized Gaussian map and the local bifurcations that govern changes in its dynamical behavior. Fixed points play a central role in discrete dynamical systems, since they correspond to steady states of the iteration and often determine the qualitative structure of nearby trajectories. Understanding their existence, stability, and bifurcations thus provides the first step toward describing the system's global dynamics.

We begin by characterizing the fixed points of the normalized map (2.2) and computing the derivative of the map at equilibrium. This allows us to determine conditions for local stability and to identify parameter values at which fixed points lose stability. In particular, we derive explicit equations describing the curves in parameter space along which saddle-node and period-doubling bifurcations occur. These bifurcations mark the transition from simple equilibrium dynamics to more complex behaviors such as periodic oscillations and cascades of period-doubling.

PROPOSITION 3.1. *Let y^* be a fixed point of T in (2.2) with $B > 0$. Then its multiplier is*

$$\eta = T'(y^*) = -2By^* e^{-(y^*)^2}.$$

Moreover, y^ is locally asymptotically stable if $|\eta| < 1$, unstable if $|\eta| > 1$, and non-hyperbolic if $|\eta| = 1$.*

Proof. A fixed point satisfies

$$y^* = T(y^*) = A + Be^{-(y^*)^2}.$$

The derivative of T is given by (2.3), hence the multiplier at y^* is

$$\eta = T'(y^*) = -2By^*e^{-(y^*)^2}.$$

The stability conclusions follow from the standard one-dimensional criterion for fixed points. $\square$

The fixed points of the normalized Gaussian map (2.2) are solutions of

$$F(y, A, B) = A + Be^{-y^2} - y = 0.$$

This equation defines a manifold in the three-dimensional space (y, A, B) representing all possible equilibria of the system. The projection of this surface onto the parameter plane determines the number and type of fixed points for each parameter pair. The saddle-node locus is obtained when the fixed-point surface loses regularity under projection onto parameter space, namely

$$F(y, A, B) = 0, \quad \partial_y F(y, A, B) = 0,$$

which is equivalent to the multiplier condition $T'(y) = 1$. A fixed-point period-doubling bifurcation is instead characterized by

$$F(y, A, B) = 0, \quad T'(y) = -1.$$

Thus, the location of the fixed point can be used as a parameter to describe the bifurcation curves in the (A, B) -plane.

THEOREM 3.2 (Fixed-point bifurcation loci and nondegeneracy). *Consider the Gaussian map T in (2.2) with $B > 0$.*

(i) *The fixed-point multiplier +1 locus is*

$$A_{\text{sn}}(y) = y + \frac{1}{2y}, \quad B_{\text{sn}}(y) = -\frac{e^{y^2}}{2y}, \quad y < 0.$$

Every point on this locus with

$$y \neq -\frac{1}{\sqrt{2}}$$

is a generic saddle-node bifurcation, with B as local unfolding parameter. The point

$$y = -\frac{1}{\sqrt{2}}, \quad (A, B) = \left(-\sqrt{2}, \sqrt{\frac{e}{2}}\right),$$

is a degenerate codimension-two cusp point of the fixed-point equation.

(ii) *The fixed-point multiplier -1 locus is*

$$A_{\text{pd}}(y) = y - \frac{1}{2y}, \quad B_{\text{pd}}(y) = \frac{e^{y^2}}{2y}, \quad y > 0.$$

Every point on this locus is a generic period-doubling (flip) bifurcation of the corresponding fixed point.

Proof. A fixed point y satisfies

$$y = A + Be^{-y^2},$$

or equivalently

$$A = y - Be^{-y^2}.$$

For a saddle-node bifurcation, the fixed point must be nonhyperbolic with multiplier $+1$, so

$$-2Bye^{-y^2} = 1.$$

Solving for B gives

$$B = -\frac{e^{y^2}}{2y}, \quad y \neq 0.$$

Since $B > 0$, this requires $y < 0$. Substituting into the fixed point equation yields

$$A = y - \left(-\frac{e^{y^2}}{2y}\right)e^{-y^2} = y + \frac{1}{2y}, \quad y < 0.$$

This gives the multiplier $+1$ locus.

For a period-doubling bifurcation, the fixed point must be nonhyperbolic with multiplier -1 , so

$$-2Bye^{-y^2} = -1.$$

Hence

$$B = \frac{e^{y^2}}{2y}, \quad y \neq 0,$$

which requires $y > 0$ because $B > 0$. Substituting into the fixed point equation gives

$$A = y - \frac{1}{2y}, \quad y > 0.$$

This gives the period-doubling curve.

It remains to verify the bifurcation nondegeneracy conditions. For the saddle-node locus, let

$$F(y, A, B) = A + Be^{-y^2} - y.$$

Along the multiplier $+1$ locus,

$$F_y = 0, \quad F_B = e^{-y^2} > 0,$$

and

$$F_{yy} = T''(y) = \frac{1 - 2y^2}{y}.$$

Hence $F_{yy} \neq 0$ except at $y = -1/\sqrt{2}$. Therefore every point of the saddle-node locus except this one is a generic fold.

At $y = -1/\sqrt{2}$,

$$F = F_y = F_{yy} = 0, \quad F_{yyy} = T'''(y) = -4 \neq 0.$$

Moreover,

$$\det \begin{pmatrix} F_A & F_B \\ F_{yA} & F_{yB} \end{pmatrix} = \sqrt{\frac{2}{e}} \neq 0.$$

Thus

$$\left(-\sqrt{2}, \sqrt{\frac{e}{2}}\right)$$

is a codimension-two cusp point of the fixed-point equation.

For the period-doubling locus, let $y = y(B)$ denote the local fixed-point branch for fixed A , and let $\eta(B) = T'(y(B))$. At the flip locus,

$$\frac{d\eta}{dB} = -\frac{e^{-y^2}(2y^2 + 1)}{2y} < 0,$$

so the multiplier crosses -1 transversally. Furthermore, the one-dimensional flip non-degeneracy coefficient is

$$\frac{1}{4}[T''(y)]^2 + \frac{1}{6}T'''(y) = \frac{y^2}{3} + \frac{1}{4y^2} > 0.$$

Hence every point of the period-doubling locus is a generic flip bifurcation. $\square$

COROLLARY 3.3 (Fixed- A bifurcation thresholds). *For the map T in (2.2), let $B > 0$ and fix $A \in \mathbb{R}$.*

(i) *Period-doubling threshold. For every $A \in \mathbb{R}$, there exists a unique period-doubling threshold, occurring at*

$$y_{\text{pd}}(A) = \frac{A + \sqrt{A^2 + 2}}{2} > 0, \quad B_{\text{pd}}(A) = \frac{e^{y_{\text{pd}}(A)^2}}{2y_{\text{pd}}(A)} > 0.$$

(ii) *Saddle-node thresholds. Saddle-node bifurcations can occur only when $A \leq -\sqrt{2}$. In that case, there are generically two saddle-node thresholds, occurring at*

$$y_{\text{sn}}^{\pm}(A) = \frac{A \pm \sqrt{A^2 - 2}}{2} < 0, \quad B_{\text{sn}}^{\pm}(A) = -\frac{e^{y_{\text{sn}}^{\pm}(A)^2}}{2y_{\text{sn}}^{\pm}(A)} > 0.$$

When $A = -\sqrt{2}$ the two roots coincide at $y_{\text{sn}}^{\pm}(-\sqrt{2}) = -1/\sqrt{2}$, giving the codimension-two cusp point of the fixed-point equation $B_{\text{sn}}(-\sqrt{2}) = \sqrt{e}/2$, which is not a generic saddle-node bifurcation. No saddle-node threshold exists for $A > -\sqrt{2}$.

Proof. Period-doubling threshold. From Theorem 3.2(ii), the period-doubling curve is parametrized by

$$A = y - \frac{1}{2y}, \quad y > 0.$$

Multiplying through by $2y > 0$ gives the quadratic

$$(3.1) \quad 2y^2 - 2Ay - 1 = 0.$$

Its discriminant is $4A^2 + 8 > 0$ for every $A \in \mathbb{R}$, so two real roots always exist. The product of the roots equals $-1/2 < 0$, so the roots have opposite signs; hence, there is exactly one positive root, namely

$$y_{\text{pd}}(A) = \frac{A + \sqrt{A^2 + 2}}{2}.$$

Substituting into the expression for B_{pd} from Theorem 3.2(ii) gives

$$B_{\text{pd}}(A) = \frac{e^{y_{\text{pd}}(A)^2}}{2y_{\text{pd}}(A)} > 0,$$

where positivity follows from $y_{\text{pd}}(A) > 0$.

Saddle-node thresholds. From Theorem 3.2(i), the saddle-node curve is parametrized by

$$A = y + \frac{1}{2y}, \quad y < 0.$$

Multiplying through by $2y < 0$ gives the quadratic

$$(3.2) \quad 2y^2 - 2Ay + 1 = 0.$$

Case 1: $|A| < \sqrt{2}$: no real roots exist, so no saddle-node threshold occurs.

Case 2: $A \geq \sqrt{2}$: two real roots exist,

$$y^{\pm} = \frac{A \pm \sqrt{A^2 - 2}}{2}.$$

Since $0 \leq \sqrt{A^2 - 2} < A$, both roots are strictly positive and therefore violate the requirement $y < 0$. Hence, no saddle-node threshold exists for $B > 0$ when $A \geq \sqrt{2}$.

Case 3: $A \leq -\sqrt{2}$: writing $A = -|A|$ with $|A| \geq \sqrt{2}$, the two roots are

$$y^{\pm} = \frac{A \pm \sqrt{A^2 - 2}}{2}.$$

Since $0 \leq \sqrt{A^2 - 2} < |A| = -A$, both quantities $A + \sqrt{A^2 - 2}$ and $A - \sqrt{A^2 - 2}$ are strictly negative, so both roots satisfy $y < 0$, consistent with the parametrization requirement. Labelling them

$$y_{\text{sn}}^+(A) = \frac{A + \sqrt{A^2 - 2}}{2}, \quad y_{\text{sn}}^-(A) = \frac{A - \sqrt{A^2 - 2}}{2},$$

we note that $y_{\text{sn}}^-(A) \leq y_{\text{sn}}^+(A) < 0$. For each root, the corresponding B value from Theorem 3.2(i) is

$$B_{\text{sn}}^{\pm}(A) = -\frac{e^{y_{\text{sn}}^{\pm}(A)^2}}{2y_{\text{sn}}^{\pm}(A)} > 0,$$

where positivity holds because $y_{\text{sn}}^{\pm}(A) < 0$. Both values correspond to genuine saddle-node bifurcations with $B > 0$, giving two distinct thresholds for $A < -\sqrt{2}$.

When $A = -\sqrt{2}$, the discriminant vanishes and the two roots coincide at

$$y_{\text{sn}}^{\pm}(-\sqrt{2}) = \frac{-\sqrt{2}}{2} = -\frac{1}{\sqrt{2}},$$

yielding the single threshold

$$B_{\text{sn}}(-\sqrt{2}) = -\frac{e^{1/2}}{2 \cdot (-1/\sqrt{2})} = \frac{\sqrt{e}\sqrt{2}}{2} = \sqrt{\frac{e}{2}}.$$

□

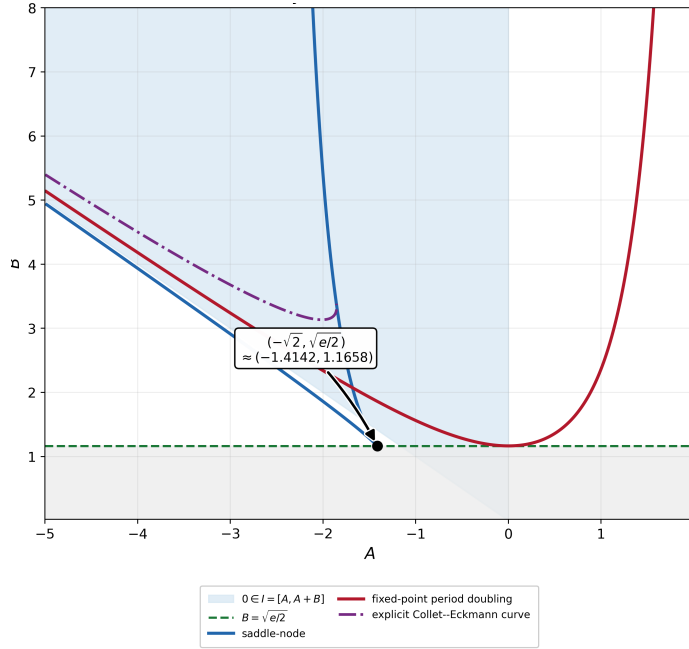


FIG. 3. *Parameter-space geometry of the normalized Gaussian family $T(y) = A + Be^{-y^2}$, with $B > 0$ showing the analytical bifurcation skeleton. The saddle-node curve is parameterized by $A_{\text{sn}}(y) = y + (2y)^{-1}$ and $B_{\text{sn}}(y) = -e^{y^2}/(2y)$ for $y < 0$, whereas the fixed-point period-doubling curve is given by $A_{\text{pd}}(y) = y - (2y)^{-1}$ and $B_{\text{pd}}(y) = e^{y^2}/(2y)$ for $y > 0$. The horizontal dashed line marks the global contraction threshold $B = \sqrt{e/2}$, and the point $(-\sqrt{2}, \sqrt{e/2})$ is the tip of the saddle-node curve. The shaded wedge corresponds to $-B \leq A \leq 0$, for which the critical point $y_{\text{cr}} = 0$ belongs to the invariant interval $I = [A, A + B]$. The explicit Collet-Eckmann parameter curve constructed in Section 4 is also displayed.*

The explicit parameterizations obtained above provide a complete analytical description of the local bifurcations of fixed points. To clarify how these curves organize the normalized parameter plane, Fig. 3 displays the saddle-node and period-doubling loci together with the global contraction threshold. The figure also indicates the region in which the critical point belongs to the invariant interval.

Figure 3 shows that the local bifurcation curves form the organizing skeleton of the parameter plane. The period-doubling curve provides the first loss-of-stability threshold of the positive fixed-point branch, while the saddle-node curve bounds the region in which three fixed points coexist. Both fixed-point bifurcation loci attain the same minimal value of the parameter B . Indeed,

$$\min_{y < 0} B_{\text{sn}}(y) = \min_{y > 0} B_{\text{pd}}(y) = \sqrt{\frac{e}{2}}.$$

For the saddle-node locus this value is attained at

$$(A, B) = \left(-\sqrt{2}, \sqrt{\frac{e}{2}} \right),$$

the cusp point, whereas for the period-doubling locus it is attained at

$$(A, B) = \left(0, \sqrt{\frac{e}{2}}\right).$$

Thus the global contraction threshold $B = \sqrt{e/2}$ coincides with the common lowest B -level at which fixed-point nonhyperbolicity can occur.

4. Global Bifurcation Structure. The results of Section 2 place the normalized Gaussian family (2.2) within the classical theory of smooth unimodal interval maps with a unique nonflat critical point and negative Schwarzian derivative [14]. In this general setting, Singer's theorem (Theorem A.7) implies that every attracting periodic orbit must attract a critical point or a boundary point [36]. Since the Gaussian family possesses a single critical point and a parameter-dependent compact invariant interval, this general theorem yields a particularly sharp consequence: whenever the critical point lies in the interior of the invariant interval, Singer's theorem (Theorem A.7), together with $T(0) = A + B$, implies that the map can support at most two attracting periodic orbits.

The results of this section go beyond this qualitative conclusion by exploiting the explicit Gaussian form. We determine the global contraction threshold, derive the first loss-of-stability mechanism, relate the attracting dynamics directly to the critical orbit, and identify explicit parameter sets exhibiting nonuniform expansion.

Throughout this section, we use standard definitions recalled in Appendix A: attracting periodic orbits (Definition A.1), basins of attraction (Definition A.2), immediate basin (Definition A.3), and ω -limit sets (Definition A.4).

THEOREM 4.1 (Number of attracting periodic orbits). *Consider T in (2.2) with $B > 0$, and let $I = [A, A + B]$. Assume that $A < 0 < A + B$, so that the unique critical point $y_{\text{cr}} = 0$ belongs to $\text{int}(I)$. Then $T : I \rightarrow I$ admits at most two attracting periodic orbits. More precisely, at most one attracting periodic orbit can have an immediate basin containing either the critical point 0 or the right endpoint $A + B$. If a second attracting periodic orbit exists, its immediate basin must contain the left endpoint A .*

Proof. From Theorem 2.1, applied to the normalized map with $\alpha = 1$, $\beta = A$, $\gamma = B$, and $\delta = 0$, we have $T(\mathbb{R}) = (A, A + B] \subset I$. Hence $T(I) \subset I$, so the dynamics may be restricted to the compact invariant interval I .

Moreover, T is C^∞ , has a unique nondegenerate quadratic critical point $y_{\text{cr}} = 0$ by Proposition 2.10, and its Schwarzian derivative satisfies $S(T)(y) < 0$ for all $y \neq 0$, by Proposition 2.11. Therefore Singer's theorem (Theorem A.7) applies to the restriction $T : I \rightarrow I$.

By Singer's theorem, the immediate basin of every attracting periodic orbit must contain either the unique critical point 0 or one of the boundary points of I , namely

$$A \quad \text{or} \quad A + B.$$

We now analyze these possibilities.

Step 1: The critical point and the right endpoint cannot correspond to two distinct attracting periodic orbits.

Since

$$T(0) = A + B,$$

if the right endpoint $A + B$ belongs to the basin of an attracting periodic orbit $\mathcal{O}$, then the critical point 0 also belongs to that basin. Indeed,

$$T^{n+1}(0) = T^n(A + B),$$

so convergence of the orbit of $A + B$ to $\mathcal{O}$ implies convergence of the orbit of 0 to the same periodic orbit.

Consequently, every attracting periodic orbit whose immediate basin contains either 0 or $A + B$ has

$$0 \in \mathcal{B}(\mathcal{O}).$$

Suppose that two distinct attracting periodic orbits $\mathcal{O}_1$ and $\mathcal{O}_2$ had this property. Then

$$0 \in \mathcal{B}(\mathcal{O}_1) \quad \text{and} \quad 0 \in \mathcal{B}(\mathcal{O}_2).$$

This is impossible because the basins of two distinct attracting periodic orbits are disjoint. Hence at most one attracting periodic orbit can have an immediate basin containing either 0 or $A + B$.

Step 2: The left endpoint can correspond to at most one additional attracting periodic orbit.

Suppose that two distinct attracting periodic orbits $\mathcal{O}_1$ and $\mathcal{O}_2$ both had immediate basins containing the left endpoint A . Then

$$A \in \mathcal{B}(\mathcal{O}_1) \quad \text{and} \quad A \in \mathcal{B}(\mathcal{O}_2),$$

which is again impossible because the basins of distinct attracting periodic orbits are disjoint.

Therefore the left endpoint A can belong to the immediate basin of at most one attracting periodic orbit.

Step 3: Conclusion.

Singer's theorem leaves only the three possibilities

$$0, \quad A, \quad A + B$$

for the critical or boundary point contained in the immediate basin of an attracting periodic orbit.

By Step 1, the points 0 and $A + B$ can account for at most one attracting periodic orbit. By Step 2, the point A can account for at most one additional attracting periodic orbit. Hence T admits at most two attracting periodic orbits.

If two attracting periodic orbits exist, one of them must therefore have the left endpoint A in its immediate basin, while the other has either the critical point 0 or the right endpoint $A + B$ in its immediate basin. In the latter case, its full basin necessarily contains the critical point 0. $\square$

COROLLARY 4.2 (Critical point attracted to a periodic orbit). *Consider T in (2.2) with $B > 0$, and let $I = [A, A + B]$. Assume that $A < 0 < A + B$, so that $y_{\text{cr}} = 0 \in \text{int}(I)$. Let $\mathcal{O} \subset I$ be an attracting periodic orbit whose immediate basin $\mathcal{B}_0(\mathcal{O})$ contains neither endpoint of I . Then $y_{\text{cr}} = 0 \in \mathcal{B}_0(\mathcal{O}) \subset \mathcal{B}(\mathcal{O})$, and $\omega(y_{\text{cr}}) = \mathcal{O}$. Moreover, $\mathcal{O}$ is the unique attracting periodic orbit whose immediate basin is disjoint from the endpoints of I .*

Proof. By Corollary 2.2, $T(\mathbb{R}) \subset I$, hence $T(I) \subset I$. By Singer's theorem (Theorem A.7), the immediate basin of every attracting periodic orbit contains either a critical point or a boundary point of I . Since $\mathcal{B}_0(\mathcal{O})$ contains neither endpoint A nor $A + B$, it must contain the unique critical point $0 \in \mathcal{B}_0(\mathcal{O})$. Therefore

$$0 \in \mathcal{B}(\mathcal{O}),$$

and consequently

$$\omega(0) = \mathcal{O}.$$

Finally, suppose that another attracting periodic orbit $\tilde{\mathcal{O}}$ also had an immediate basin disjoint from the endpoints of I . The same argument would give

$$0 \in \mathcal{B}(\tilde{\mathcal{O}}),$$

which is impossible because the basins of two distinct attracting periodic orbits are disjoint. Hence $\mathcal{O}$ is the unique attracting periodic orbit whose immediate basin is disjoint from the endpoints of I . $\square$

This corollary shows that, whenever an attracting periodic orbit has an immediate basin disjoint from the endpoints of I , it is completely determined by the asymptotic behavior of the critical orbit. In this sense, the forward iterates of the critical point organize the attracting dynamics of the normalized Gaussian family (2.2). This provides a practical criterion for numerical exploration of attracting periodic orbits whose immediate basins avoid the endpoints, since such orbits are detected by following the forward orbit of the unique critical point.

Figure 4 illustrates this mechanism for representative parameter values along the slice $A = -1$. The four panels show the evolution of the critical orbit as the parameter B increases through successive dynamical regimes. Convergence to an attracting fixed point is followed by attracting periodic solutions of increasing period before the onset of an irregular expanding regime. The figure therefore provides a direct visual interpretation of Corollary 4.2, emphasizing that the system's asymptotic behavior of the critical orbit is encoded in its forward iterates.

The temporal evolution presented in Fig. 4 describes how the critical orbit approaches the attracting invariant set for representative parameter values. The classical cobweb construction provides an equivalent geometric interpretation, showing how successive iterations alternate between the graph of the map and the identity line.

Figure 5 illustrates this mechanism for the same parameter values considered previously. The four panels exhibit convergence toward an attracting fixed point, an attracting period-two orbit, an attracting period-four orbit, and a numerically irregular trajectory. The transition between these regimes reflects the progressive loss of stability of the attracting fixed point through the period-doubling cascade observed numerically after the fixed-point flip bifurcation identified analytically in the previous section.

THEOREM 4.3 (Fixed point and global attraction). *Consider T in (2.2) with $B > 0$ and fixed $A \in \mathbb{R}$. Then the following hold:*

- (i) *For every $0 < B < \sqrt{e/2}$, the map T is a contraction on $\mathbb{R}$ and admits a unique fixed point $y_B^* \in \mathbb{R}$. In particular, the fixed point y_B^* is globally attracting, and for every $y_0 \in \mathbb{R}$ the orbit defined by*

$$y_{n+1} = T(y_n)$$

satisfies

$$y_n \rightarrow y_B^* \quad \text{as } n \rightarrow \infty.$$

- (ii) *For any $B > 0$, the map T admits a unique fixed point $y_B^* \in \mathbb{R}$ if $|A| < \sqrt{2}$.*

Proof. Step 1: Contraction bound. From the differentiation of T in (2.3) we have

$$|T'(y)| = 2B|y|e^{-y^2}.$$

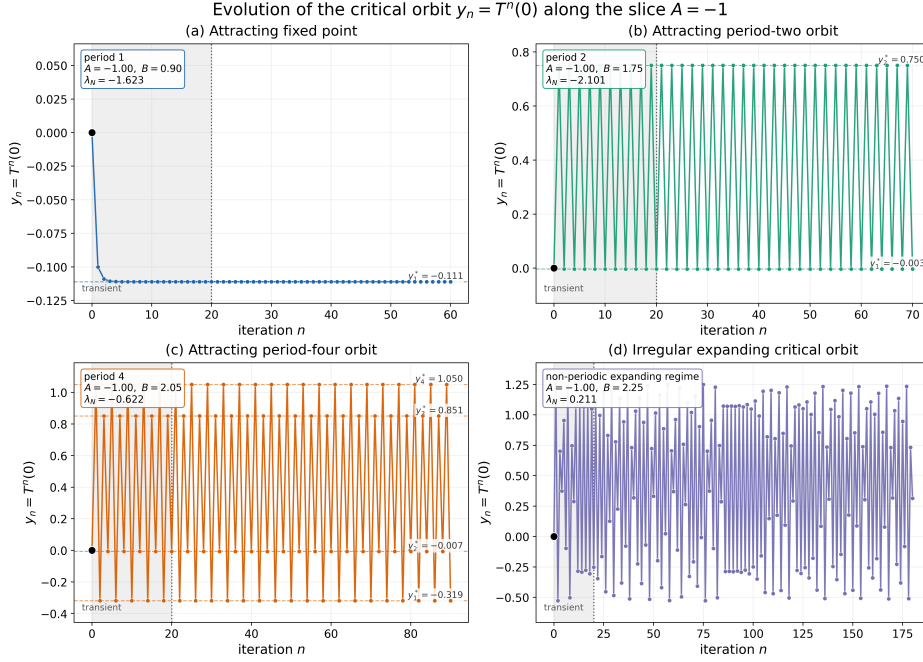


FIG. 4. Evolution of the critical orbit $c_n = T^n(0)$ for the normalized Gaussian map $T(y) = A + Be^{-y^2}$ along the parameter slice $A = -1$. Panel (a) shows convergence to an attracting fixed point. Panel (b) shows locking onto an attracting period-two orbit, whereas panel (c) displays convergence to an attracting period-four orbit. Panel (d) illustrates a numerically irregular regime with a positive orbit-averaged expansion rate. The shaded initial portion of each panel identifies the transient interval discarded when estimating the asymptotic behavior. In the periodic cases, the dashed horizontal lines indicate the numerically detected asymptotic orbit levels. These examples illustrate how the critical orbit successively detects the attracting regimes encountered along the parameter slice and thus provides a natural probe of the global dynamics.

To estimate its supremum, consider

$$g(y) := |y|e^{-y^2}.$$

By symmetry it suffices to maximize g on $[0, \infty)$. For $y > 0$,

$$g'(y) = e^{-y^2}(1 - 2y^2),$$

so the unique critical point on $(0, \infty)$ is

$$y = \frac{1}{\sqrt{2}},$$

and this point is a global maximum. Therefore

$$\sup_{y \in \mathbb{R}} |T'(y)| = 2B \left(\frac{1}{\sqrt{2}} \right) e^{-1/2} = B\sqrt{2}e^{-1/2}.$$

If

$$B < \sqrt{\frac{e}{2}},$$

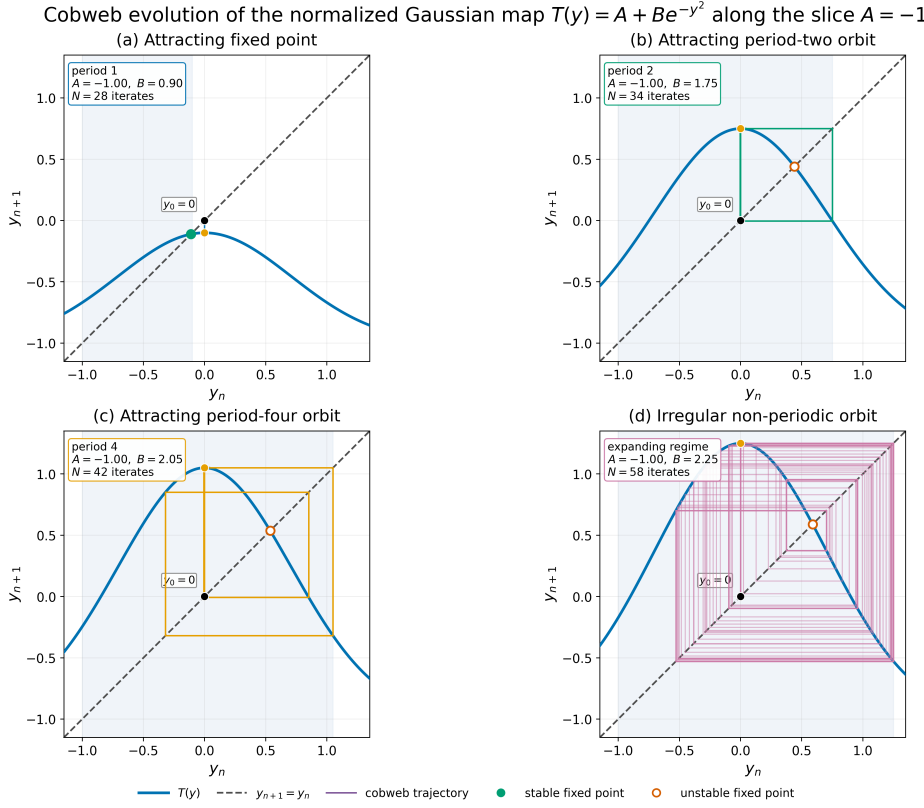


FIG. 5. Cobweb evolution of the normalized Gaussian map $T(y) = A + Be^{-y^2}$ along the parameter slice $A = -1$. Panels (a)–(d) correspond respectively to an attracting fixed point, an attracting period-two orbit, an attracting period-four orbit, and an irregular non-periodic trajectory obtained after the loss of low-period stability. The blue curve represents the Gaussian map, the dashed line is the identity map, and the colored cobweb paths depict successive iterations starting from the critical point $y_0 = 0$. The shaded region indicates the invariant interval $I = [A, A + B]$. The figure provides a geometric interpretation of the orbit evolution shown in Fig. 4 and illustrates how the nonlinear folding mechanism generates increasingly complex asymptotic dynamics as the parameter B increases.

then

$$B\sqrt{2}e^{-1/2} < 1,$$

and hence T is a contraction on $\mathbb{R}$.

Step 2: Global attraction. Since the metric space $(\mathbb{R}, |\cdot|)$ is complete and T is a contraction, the Banach Fixed point Theorem (see Theorem A.6 in the appendix) implies that T has a unique fixed point and that, for every initial condition $y_0 \in \mathbb{R}$, the iterates

$$y_{n+1} = T(y_n)$$

converge to this fixed point. This proves (i).

Step 3: Fixed point parametrization. A $y \in \mathbb{R}$ is a fixed point of T if and only if

$$y = A + Be^{-y^2},$$

or equivalently,

$$B = e^{y^2}(y - A).$$

Thus, the fixed points are parameterized by the function

$$\Phi_A(y) := e^{y^2}(y - A).$$

Step 4: Monotonicity of Φ_A . Differentiating, we obtain

$$\Phi'_A(y) = e^{y^2}(1 + 2y(y - A)) = e^{y^2}(2y^2 - 2Ay + 1).$$

Since $e^{y^2} > 0$ for all $y \in \mathbb{R}$, the sign of $\Phi'_A(y)$ is the sign of the quadratic polynomial

$$q_A(y) := 2y^2 - 2Ay + 1.$$

Its discriminant is

$$\Delta_A = (-2A)^2 - 8 = 4A^2 - 8.$$

For $|A| < \sqrt{2}$, we have $\Delta_A < 0$. Since the leading coefficient of q_A is positive, it follows that

$$q_A(y) > 0 \quad \text{for all } y \in \mathbb{R}.$$

Hence

$$\Phi'_A(y) > 0 \quad \text{for all } y \in \mathbb{R},$$

so Φ_A is strictly increasing on $\mathbb{R}$ whenever $|A| < \sqrt{2}$.

Step 5: Surjectivity of Φ_A . Here we compute the limits of $\Phi_A(y)$ at $y \rightarrow \pm\infty$. As $y \rightarrow +\infty$: both $e^{y^2} \rightarrow +\infty$ and $y - A \rightarrow +\infty$, so $\Phi_A(y) \rightarrow +\infty$. On the other side, for $y \rightarrow -\infty$, write $y = A - t$ with $t \rightarrow +\infty$, so that

$$\Phi_A(y) = e^{(A-t)^2}(-t).$$

Since $e^{(A-t)^2} > 0$ and $-t \rightarrow -\infty$, the product $\Phi_A(y) \rightarrow -\infty$.

Step 6: Existence and uniqueness. Observe that Φ_A is continuous, strictly increasing for $|A| < \sqrt{2}$, and satisfies $\Phi_A(y) \rightarrow \pm\infty$ as $y \rightarrow \pm\infty$, so it is a bijection from $\mathbb{R} \rightarrow \mathbb{R}$. For any $B > 0$, the equation $\Phi_A(y) = B$ therefore has a unique solution $y_B^* \in \mathbb{R}$, which is the unique fixed point of T . This proves (ii). $\square$

The two parts of Theorem 4.3 are independent: part (i) establishes global attraction for B sufficiently small with no condition on A , while part (ii) guarantees uniqueness of the fixed point for all $B > 0$ under a condition on A .

The bifurcation curves of Theorem 3.2 and Corollary 3.3 can now be assembled into a complete description of the (A, B) -parameter plane.

THEOREM 4.4 (Parameter-plane structure). *Consider T in (2.2) with $B > 0$. Then the following hold:*

- (i) *If $A > -\sqrt{2}$, then T has exactly one fixed point for every $B > 0$.*
- (ii) *If $A = -\sqrt{2}$, then T has exactly one fixed point for every $B > 0$, and this fixed point is nonhyperbolic at*

$$B = \sqrt{\frac{e}{2}}.$$

(iii) If $A < -\sqrt{2}$, then T has three fixed points for

$$B \in (B_{\text{sn}}^+(A), B_{\text{sn}}^-(A)),$$

two fixed points when $B = B_{\text{sn}}^\pm(A)$, and one outside this interval.

Proof. Fixed points satisfy

$$\Phi_A(y) := e^{y^2}(y - A) = B.$$

Since $B > 0$, every fixed point must satisfy

$$y - A > 0, \quad \text{i.e.} \quad y > A.$$

Thus it is enough to analyze Φ_A behavior on the interval (A, ∞) by means of the sign of $\Phi'_A(y)$, which is given by the sign of $q_A(y) = 2y^2 - 2Ay + 1$.

Case 1: $A > -\sqrt{2}$. Since q_A is a convex quadratic, its minimum over (A, ∞) occurs either at the vertex $y = A/2$ or at the endpoint $y = A$.

If $A > 0$, then $A/2 < A$, so the minimum on (A, ∞) is attained at $y = A$, and

$$q_A(A) = 1 > 0.$$

If $-\sqrt{2} < A \leq 0$, then $A/2 \geq A$, so the minimum occurs at $y = A/2$, and

$$q_A\left(\frac{A}{2}\right) = 1 - \frac{A^2}{2} > 0.$$

Therefore $\Phi'_A(y) > 0$ for all $y > A$, so Φ_A is strictly increasing on (A, ∞) . Since

$$\lim_{y \rightarrow A^+} \Phi_A(y) = 0, \quad \lim_{y \rightarrow +\infty} \Phi_A(y) = +\infty,$$

the equation $\Phi_A(y) = B$ admits exactly one solution for every $B > 0$.

Case 2: $A = -\sqrt{2}$. In this case,

$$q_A(y) = 2y^2 + 2\sqrt{2}y + 1 = (\sqrt{2}y + 1)^2 \geq 0,$$

with equality only at

$$y = -\frac{1}{\sqrt{2}}.$$

Hence Φ_A is nondecreasing on (A, ∞) , strictly increasing except at that single point, and still satisfies

$$\lim_{y \rightarrow A^+} \Phi_A(y) = 0, \quad \lim_{y \rightarrow +\infty} \Phi_A(y) = +\infty.$$

Thus $\Phi_A(y) = B$ has exactly one solution for every $B > 0$. At $y = -\frac{1}{\sqrt{2}}$, one finds

$$B = \Phi_A\left(-\frac{1}{\sqrt{2}}\right) = \sqrt{\frac{e}{2}},$$

and the multiplier is

$$\eta = T'(-1/\sqrt{2}) = -2\sqrt{\frac{e}{2}} \left(\frac{-1}{\sqrt{2}}\right) e^{-1/2} = 1,$$

so the fixed point is nonhyperbolic there.

Case 3: $A < -\sqrt{2}$. Now the quadratic q_A has two distinct real roots, namely

$$y_{\text{sn}}^{\pm}(A) = \frac{A \pm \sqrt{A^2 - 2}}{2},$$

and both lie in $(A, 0)$. Therefore Φ_A has a local maximum and a local minimum at these two points, respectively. Their corresponding values

$$B_{\text{sn}}^{\pm}(A) = \Phi_A(y_{\text{sn}}^{\pm}(A))$$

are precisely the saddle-node values. Hence the horizontal line $B = \text{const}$ intersects the graph of Φ_A three times when

$$B \in (B_{\text{sn}}^{+}(A), B_{\text{sn}}^{-}(A)),$$

twice at the endpoints, and once outside this interval. This gives the claimed fixed point count. $\square$

The preceding results describe the organization of attracting periodic dynamics. But the Gaussian map family (2.2) also contains explicit parameter values for which the critical orbit exhibits exponential derivative growth. This provides a concrete mechanism for the emergence of chaotic statistical regimes within the family. We now introduce a standard condition that provides a mechanism for the onset of chaotic behavior in unimodal maps.

Theorem 3.2, Corollary 3.3, and Theorem 4.4 determine the regions in which the normalized Gaussian map possesses one or three fixed points and identify the saddle-node and fixed-point period-doubling bifurcation curves. These local bifurcations constitute only the first stage of the global dynamical organization.

To visualize the subsequent evolution of the attracting invariant set, the Fig. 6 presents one-parameter bifurcation diagrams obtained by fixing the parameter A and varying B . For each parameter value, the critical orbit is iterated, an initial transient is discarded, and the remaining iterates are plotted. The analytical bifurcation thresholds derived previously are superimposed, allowing a direct comparison between the theoretical predictions and the observed asymptotic dynamics.

Figure 6 illustrates that the analytical bifurcation curves derived in Section 3 accurately predict the onset of the first qualitative changes in the dynamics. For each parameter slice, the fixed-point period-doubling threshold marks the analytically predicted loss of stability of the attracting equilibrium and is followed numerically by a period-doubling cascade.

The diagrams also emphasize the influence of the offset parameter A . For $A = -3$, nontrivial dynamics are confined to a relatively narrow interval of B , whereas increasing A substantially enlarges the parameter region supporting higher-period attractors and chaotic behavior. The resulting sequence of bifurcations closely follows the universal scenario observed in smooth unimodal maps, while retaining the distinctive global geometry imposed by the Gaussian nonlinearity.

DEFINITION 4.5 (Collet-Eckmann condition). *Let $T : I \rightarrow I$ be a smooth unimodal map with critical point $y_{\text{cr}} \in I$. Denote by*

$$y_0 = T(y_{\text{cr}}), \quad y_{k+1} = T(y_k)$$

the post-critical orbit. The map T is said to satisfy the Collet-Eckmann condition if there exist constants $C > 0$ and $\Lambda > 1$ such that

$$|(T^n)'(y_0)| \geq C\Lambda^n, \quad n \geq 1.$$

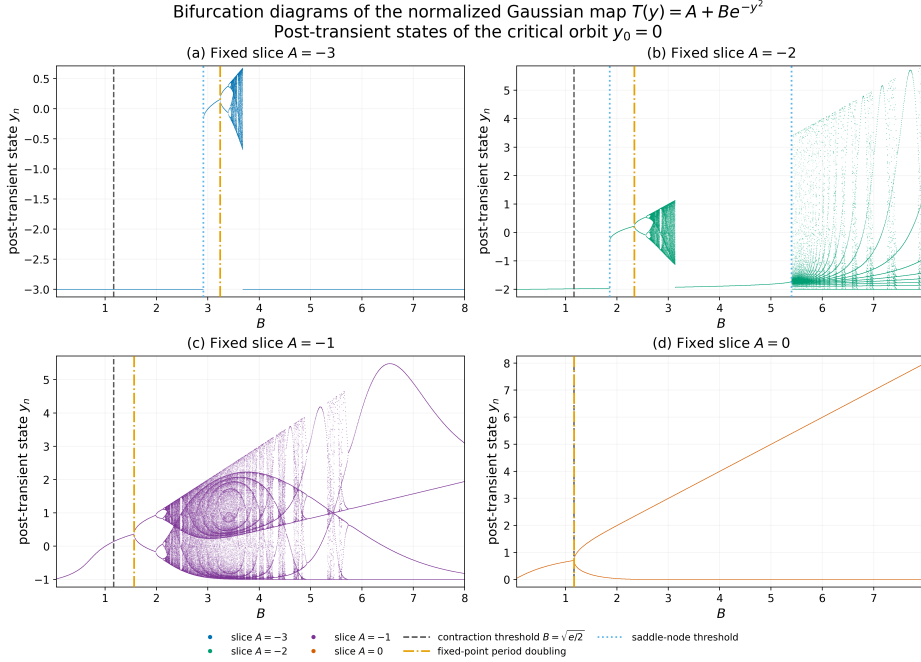


FIG. 6. Bifurcation diagrams of the normalized Gaussian map $T(y) = A + Be^{-y^2}$ for four representative parameter slices: (a) $A = -3$, (b) $A = -2$, (c) $A = -1$, and (d) $A = 0$. For each value of B , the critical orbit $y_0 = 0$ was iterated, an initial transient was discarded, and the remaining orbit points were plotted. The dashed, dash-dotted, and dotted vertical lines indicate, respectively, the global contraction threshold $B = \sqrt{e/2}$, the fixed-point period-doubling threshold, and the saddle-node bifurcation thresholds predicted analytically. The diagrams show the transition from stable fixed points to period-doubling cascades, periodic windows, and apparently chaotic regimes, illustrating how the analytically derived local bifurcations organize the global dynamics.

The Collet-Eckmann condition is a standard quantitative form of nonuniform hyperbolicity for unimodal maps. It requires exponential derivative growth along the forward orbit of the critical value [12, 26]. Under suitable recurrence and regularity assumptions, it implies the existence of an absolutely continuous invariant probability measure and strong statistical properties [2, 22].

In the present setting, we do not reprove the general Collet-Eckmann theory. Instead, we exploit the Gaussian structure to construct parameter values for which the post-critical orbit and its derivative growth are computed analytically. By showing that the normalized Gaussian family (2.2) contains explicit parameter values for which this condition is satisfied, we provide a concrete realization of nonuniformly expanding dynamics within the family.

PROPOSITION 4.6 (An explicit post-critical orbit). *Let $q > 0$, and define*

$$A(q) = -\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}}, \quad B(q) = \frac{2q}{1 - e^{-q^2}}.$$

Then, for the map

$$T_q(y) = A(q) + B(q)e^{-y^2},$$

the critical orbit satisfies

$$0 \mapsto q \mapsto -q \mapsto -q.$$

846 *In particular, the critical point is preperiodic and lands on the fixed point $-q$.*

847 *Proof.* Since the critical point of T_q is $y_{\text{cr}} = 0$, we first compute

$$848 \quad T_q(0) = A(q) + B(q).$$

849 Using the definitions of $A(q)$ and $B(q)$, we obtain

$$850 \quad A(q) + B(q) = -\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}} + \frac{2q}{1 - e^{-q^2}} = q.$$

851 Thus $T_q(0) = q$. Next,

$$852 \quad T_q(q) = A(q) + B(q)e^{-q^2}.$$

853 Substituting the expressions for $A(q)$ and $B(q)$ gives

$$854 \quad T_q(q) = -\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}} + \frac{2qe^{-q^2}}{1 - e^{-q^2}} = -\frac{q(1 - e^{-q^2})}{1 - e^{-q^2}} = -q.$$

855 Since T_q is an even function apart from the additive constant, the same computation
856 gives

$$857 \quad T_q(-q) = A(q) + B(q)e^{-q^2} = -q.$$

858 Therefore,

$$859 \quad 0 \mapsto q \mapsto -q \mapsto -q,$$

860 as claimed. □

861 **THEOREM 4.7** (Explicit Collet-Eckmann parameters). *Let $q_* > 0$ be the unique*
862 *positive solution of*

$$863 \quad e^{q_*^2} = 1 + 4q_*^2.$$

864 *For every $q \in (0, q_*)$, the map*

$$865 \quad T_q(y) = A(q) + B(q)e^{-y^2},$$

866 *with*

$$867 \quad A(q) = -\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}}, \quad B(q) = \frac{2q}{1 - e^{-q^2}},$$

868 *satisfies the Collet-Eckmann condition. More precisely, if*

$$869 \quad y_0 = T_q(0), \quad y_{k+1} = T_q(y_k),$$

870 *then there exists $\Lambda(q) > 1$ such that*

$$871 \quad |(T_q^n)'(y_0)| \geq \Lambda(q)^n, \quad n \geq 1.$$

872 *Proof.* By Proposition 4.6,

$$873 \quad y_0 = q, \quad y_k = -q \quad \text{for every } k \geq 1.$$

874 Since

$$875 \quad T_q'(y) = -2B(q)ye^{-y^2},$$

876 we have

$$877 \quad |T_q'(q)| = |T_q'(-q)| = 2B(q)qe^{-q^2}.$$

Using the expression for $B(q)$, this becomes

$$2B(q)qe^{-q^2} = 2 \left(\frac{2q}{1 - e^{-q^2}} \right) qe^{-q^2} = \frac{4q^2 e^{-q^2}}{1 - e^{-q^2}} = \frac{4q^2}{e^{q^2} - 1}.$$

Define

$$\Lambda(q) = \frac{4q^2}{e^{q^2} - 1}.$$

Then

$$|(T_q^n)'(y_0)| = |T_q'(q)| |T_q'(-q)|^{n-1} = \Lambda(q)^n.$$

It remains to identify when $\Lambda(q) > 1$. This is equivalent to

$$\frac{4q^2}{e^{q^2} - 1} > 1,$$

or

$$e^{q^2} < 1 + 4q^2.$$

Let

$$h(q) = 1 + 4q^2 - e^{q^2},$$

so that

$$h'(q) = 2q(4 - e^{q^2}).$$

Note that h is strictly increasing on $0 < q < \sqrt{\log 4}$ and strictly decreasing on $q > \sqrt{\log 4}$. Since $h(0) = 0$ and $h(q) \rightarrow -\infty$ as $q \rightarrow +\infty$, there is exactly one zero $q_* > 0$, defined by

$$e^{q_*^2} = 1 + 4q_*^2.$$

Thus, for every $q \in (0, q_*)$,

$$\Lambda(q) > 1.$$

Consequently,

$$|(T_q^n)'(y_0)| \geq \Lambda(q)^n, \quad n \geq 1,$$

which is the Collet-Eckmann condition with $C = 1$. $\square$

Theorem 4.7 establishes the existence of an explicit one-parameter family of Collet-Eckmann maps. Although the proof is entirely analytical, it is instructive to compare this family with the global numerical organization of the parameter plane. Figure 7 presents orbit-averaged expansion rate

$$\lambda_N = \frac{1}{N} \sum_{k=1}^N \log |T'(y_k)|,$$

computed along the critical orbit $y_0 = 0$ over the (A, B) parameter space. The explicit Collet-Eckmann curve derived above is superimposed on the numerical diagram, allowing a direct comparison between the rigorous analytical construction and the surrounding dynamical regimes.

Moreover, for these parameters, the map cannot possess an attracting periodic orbit whose immediate basin is disjoint from the endpoints of I . Indeed, Corollary 4.2 would imply that such an orbit attracts the critical point, whereas the critical orbit lands on the repelling fixed point $-q$.

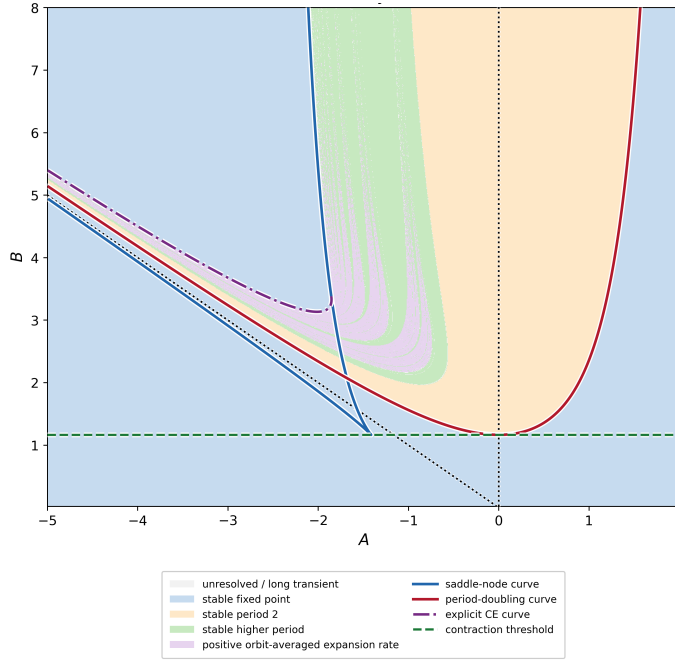


FIG. 7. *Parameter-space distribution of the orbit-averaged expansion rate computed along the critical orbit $y_0 = 0$ of the normalized Gaussian map $T(y) = A + Be^{-y^2}$. Positive values indicate average local expansion along the finite critical orbit, whereas negative values indicate net contraction along the sampled finite orbit segment. We superimpose the analytical saddle-node, period-doubling, contraction, and explicit Collet-Eckmann curves using the same color convention adopted throughout the paper. The explicit Collet-Eckmann family lies entirely inside the expanding region, in agreement with Theorem 4.7. Although the numerical map is computed from finite trajectories, it provides a global overview of the parameter-space organization and complements the rigorous analytical bifurcation analysis.*

Theorem 4.7 shows that the Collet-Eckmann regime is not merely a formal possibility for the Gaussian family (2.2). It occurs along the explicit parameter curve

$$\Gamma_{\text{CE}} = \left\{ \left(-\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}}, \frac{2q}{1 - e^{-q^2}} \right) : 0 < q < q_* \right\}.$$

Along this curve, the critical orbit lands after two iterations on the repelling fixed point $-q$. Hence, the post-critical orbit is explicitly computable, and the exponential derivative growth required by the Collet-Eckmann condition follows directly.

This curve lies in the non-contractive parameter region. Indeed, as q varies in $(0, q_*)$, the corresponding values of $B(q)$ satisfy

$$B(q) > \sqrt{\frac{e}{2}},$$

which is consistent with the fact that Collet-Eckmann behavior cannot occur in the global contraction regime.

Remark 4.8 (Gaussian expansion threshold). The value q_* defined in Theorem 4.7 can be expressed in closed form using the Lambert W function:

$$q_* = \sqrt{-W_{-1}\left(-\frac{e^{-1/4}}{4}\right) - \frac{1}{4}} = 1.52861472656227 \dots$$

We now combine the explicit Collet-Eckmann construction above with the classical statistical theory of unimodal maps. The following result records the consequence of Collet-Eckmann growth for the normalized Gaussian family (2.2).

In the next theorem, the notation $T_*\mu = \mu$ means that the measure μ is invariant under the map T . More precisely, for every measurable set $E \subset I$,

$$\mu(T^{-1}(E)) = \mu(E).$$

Equivalently, for every integrable observable $\varphi : I \rightarrow \mathbb{R}$,

$$\int_I \varphi(T(y)) d\mu(y) = \int_I \varphi(y) d\mu(y).$$

Thus, the statistical distribution of states is unchanged by the dynamics.

THEOREM 4.9 (Statistical consequence of Collet-Eckmann behavior). *For the normalized Gaussian map T in (2.2) with $B > 0$ and $I = [A, A + B]$, assume that $A < 0 < A + B$, so that the unique critical point $y_{\text{cr}} = 0 \in \text{int}(I)$. Suppose that T satisfies the Collet-Eckmann condition, namely, that there exist constants $C > 0$ and $\Lambda > 1$ such that*

$$|(T^n)'(T(y_{\text{cr}}))| \geq C\Lambda^n, \quad n \geq 1.$$

Then $T : I \rightarrow I$ admits an ergodic absolutely continuous invariant probability measure μ , i.e.,

$$T_*\mu = \mu, \quad d\mu = \rho(y) dy,$$

where

$$\rho \in L^1(I), \quad \rho \geq 0, \quad \int_I \rho(y) dy = 1.$$

Moreover, μ is a physical measure: there exists a measurable set $B(\mu) \subset I$ of positive Lebesgue measure such that, for every $y_0 \in B(\mu)$ and every continuous observable $\varphi : I \rightarrow \mathbb{R}$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^{N-1} \varphi(T^k(y_0)) = \int_I \varphi d\mu.$$

Proof. By Corollary 2.2, the interval $I = [A, A + B]$ is forward invariant, so the dynamics may be restricted to $T : I \rightarrow I$.

From Proposition 2.10, the map T has a unique critical point $y_{\text{cr}} = 0$, and this critical point is nondegenerate and quadratic. From Proposition 2.11, the Schwarzian derivative of T satisfies

$$S(T)(y) < 0 \quad \text{for all } y \neq 0.$$

Therefore, the restriction $T : I \rightarrow I$ satisfies the regularity and unimodality assumptions of Theorem A.8. Since the Collet-Eckmann condition is assumed, that theorem yields an ergodic absolutely continuous invariant probability measure

$$d\mu = \rho(y) dy.$$

It remains to establish the physicality statement. By the Birkhoff ergodic theorem, for every $\varphi \in C(I)$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^{N-1} \varphi(T^k(y)) = \int_I \varphi d\mu$$

for μ -almost every $y \in I$. Since $C(I)$ is separable, there exists a measurable set $B(\mu) \subset I$ with

$$\mu(B(\mu)) = 1$$

on which this convergence holds simultaneously for every $\varphi \in C(I)$.

Finally, since μ is absolutely continuous with respect to Lebesgue measure,

$$\mu(B(\mu)) = 1$$

implies

$$\text{Leb}(B(\mu)) > 0.$$

Therefore μ is physical. $\square$

Theorem 4.9 provides an ergodic absolutely continuous invariant probability measure μ with a basin of positive Lebesgue measure. The conclusion is therefore basin-dependent: it describes trajectories belonging to $B(\mu)$, and does not in general exclude the coexistence of another attractor associated with a boundary of the invariant interval.

Therefore, in this parameter regime, the Gaussian map admits a rigorous statistical description on the physical basin $B(\mu)$. For every $y_0 \in B(\mu)$ and every continuous observable $\varphi : I \rightarrow \mathbb{R}$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^{N-1} \varphi(T^k(y_0)) = \int_I \varphi d\mu.$$

In particular, trajectories in $B(\mu)$ cannot be asymptotic to a periodic attractor, since in that case their empirical measures would converge to an atomic invariant measure supported on the corresponding periodic orbit, rather than to the absolutely continuous measure μ .

Remark 4.10. The Collet-Eckmann condition concerns the exponential derivative growth along the post-critical orbit and does not, by itself, exclude the coexistence of an attracting periodic orbit whose immediate basin contains a boundary point of the invariant interval. Accordingly, the statistical conclusions above are basin-dependent: they hold for μ -almost every point and, in particular, on the physical basin $B(\mu)$, but they do not provide a global description of all trajectories in I .

The results above therefore show that the normalized Gaussian family (2.2) contains both simple attracting regimes and explicit Collet-Eckmann regimes. In the latter case, an ergodic absolutely continuous invariant probability measure governs the long-time statistics of a set of initial conditions of positive Lebesgue measure. This motivates the statistical analysis of the next section, where we derive the Perron-Frobenius operator of the Gaussian map in closed form and use it to study the local structure of invariant densities.

5. Statistical Behavior. Transfer operators provide the standard linear framework for describing the evolution of probability densities under nonlinear maps. For piecewise monotone expanding transformations, the classical work of Lasota and Yorke established the existence of absolutely continuous invariant measures through bounded-variation estimates [23]. This approach was extended to broader classes of interval transformations and developed into a spectral theory of Perron-Frobenius operators [2, 9, 20, 32].

The normalized Gaussian map (2.2) is not uniformly expanding, because its derivative vanishes at the critical point. Consequently, when an absolutely continuous invariant measure exists, its density may be unbounded and may develop singularities at the critical value and its forward images, as is typical for smooth unimodal systems. What distinguishes the Gaussian family is that both inverse branches are explicit. We can therefore derive the transfer operator without an implicit inversion step and use its denominator to identify the local singularity mechanism directly.

Given a map T , its statistical analysis is done with the aid of the Perron-Frobenius operator, which is defined as

$$(5.1) \quad (\mathcal{L}_T \rho)(z) = \sum_{y \in T^{-1}(z)} \frac{\rho(y)}{|T'(y)|},$$

whenever the preimages are real, and the derivative of T does not vanish at those preimages. This operator describes the evolution of probability densities under the action of the map T . If ρ_n denotes the density of a distribution of initial conditions evolved by T after n iterations, then

$$(5.2) \quad \rho_{n+1} = \mathcal{L}_T \rho_n.$$

In particular, invariant densities are precisely the fixed points of $\mathcal{L}_T$, that is, the solutions of

$$(5.3) \quad \mathcal{L}_T \rho = \rho.$$

While the map T acts nonlinearly on points, its action on densities is linear, and this is the appropriate framework for studying invariant densities, physical measures, and the formation of singularities in the long-time distribution of orbits. In particular, once the Perron-Frobenius operator is known explicitly, one can derive the functional equation satisfied by every invariant density and analyze its local asymptotic structure near the critical value and its forward images. In what follows we derive this operator in a closed form format.

Remark 5.1. For the derivation of the transfer operator, we first regard $T : \mathbb{R} \rightarrow \mathbb{R}$ and let $\mathcal{L}_T$ act on $L^1(\mathbb{R})$. Whenever ρ is an invariant density for the restricted dynamical system $T : I \rightarrow I$, with $I = [A, A + B]$, we identify ρ with its extension by zero outside I . With this convention, the transfer-operator formula may be written using all real preimages of z , without additional indicator functions.

THEOREM 5.2 (Perron-Frobenius operator for the Gaussian map). *Consider T in (2.2) with $B > 0$, and let $\rho \in L^1(\mathbb{R})$. Then the Perron-Frobenius operator $\mathcal{L}_T$ associated with T is given by*

$$(5.4) \quad (\mathcal{L}_T \rho)(z) = \frac{\rho\left(\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right) + \rho\left(-\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right)}{2|z-A|\sqrt{-\ln\left(\frac{z-A}{B}\right)}},$$

for almost every $z \in (A, A + B)$. For almost every $z \notin [A, A + B]$,

$$(5.5) \quad (\mathcal{L}_T \rho)(z) = 0.$$

1038

1039 *Proof.* For a one-dimensional C^1 map, the Perron-Frobenius operator acts on
1040 densities by the standard formula (5.1), where the sum is taken over all real preimages
1041 of z under T . We therefore start by explicitly determining the real preimages of a
1042 given point z .

1043 Solving $T(y) = z$ yields

$$1044 \quad \frac{z - A}{B} = e^{-y^2}.$$

1045 Since $e^{-y^2} \in (0, 1]$ for all $y \in \mathbb{R}$, real preimages exist if and only if

$$1046 \quad 0 < \frac{z - A}{B} \leq 1.$$

1047 Taking logarithms, we obtain

$$1048 \quad -y^2 = \ln\left(\frac{z - A}{B}\right),$$

1049 hence

$$1050 \quad y^2 = -\ln\left(\frac{z - A}{B}\right).$$

1051 Therefore, the two inverse branches are

$$1052 \quad (5.6) \quad y_{\pm}(z) = \pm \sqrt{-\ln\left(\frac{z - A}{B}\right)}.$$

1053 Equation (5.6) shows that every admissible point $z \in (A, A + B)$ possesses exactly
1054 two preimages, one on each monotone branch of the Gaussian map. This two-to-one
1055 geometry is the fundamental mechanism responsible for the structure of the Perron-
1056 Frobenius operator, since the probability density at z receives contributions from both
1057 inverse branches.

1058 Figure 8 illustrates this geometric construction. Panel (a) shows how a hori-
1059 zontal level intersects the left and right monotone branches of the Gaussian map,
1060 while panel (b) displays the explicit inverse branches $y_-(z)$ and $y_+(z)$, which merge
1061 continuously at the critical value $z = A + B$.

1062 If $y \in T^{-1}(z)$, then

$$1063 \quad Be^{-y^2} = z - A.$$

1064 Substituting this into (2.3) yields

$$1065 \quad T'(y) = -2y(z - A).$$

1066 Hence

$$1067 \quad |T'(y)| = 2|y||z - A|.$$

1068 Applying (5.6), we obtain

$$1069 \quad (5.7) \quad |T'(y_{\pm}(z))| = 2|z - A| \sqrt{-\ln\left(\frac{z - A}{B}\right)}.$$

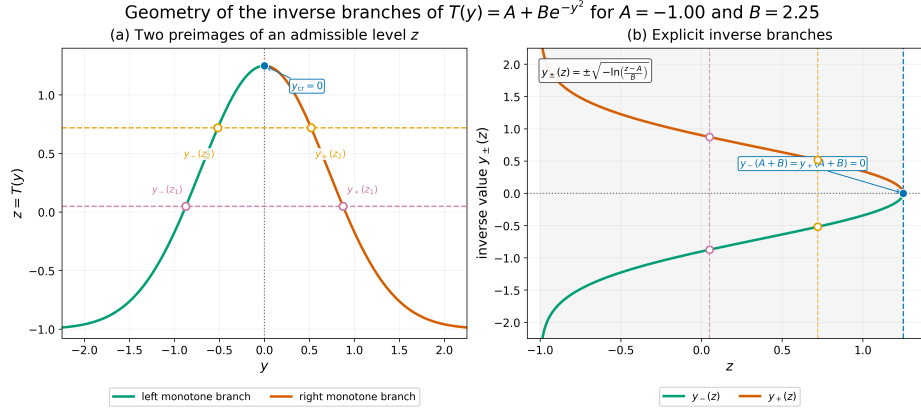


FIG. 8. Geometry of the inverse branches of the normalized Gaussian map $T(y) = A + Be^{-y^2}$. (a) Every admissible level $z \in (A, A + B)$ intersects the graph of the map at exactly two points, corresponding to the left and right monotone branches. These intersections define the two inverse images $y_-(z)$ and $y_+(z)$. (b) Explicit representation of the inverse branches $y_{\pm}(z)$ defined for $A < z < A + B$. Both branches meet continuously at the critical value $z = A + B$, where $y_-(A + B) = y_+(A + B) = 0$. This two-branch geometry provides the geometric basis for the Perron-Frobenius operator, whose action is obtained by summing the contributions transported through the two inverse branches.

1070 This quantity is the same for both branches.

1071 Substituting (5.6) and (5.7) into (5.1), we obtain

$$1072 \quad (\mathcal{L}_T \rho)(z) = \frac{\rho\left(\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right) + \rho\left(-\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right)}{2|z - A|\sqrt{-\ln\left(\frac{z-A}{B}\right)}},$$

1073 which is exactly (5.4).

1074 Finally, if

$$1075 \quad 0 < \frac{z - A}{B} \leq 1$$

1076 fails, then the equation $T(y) = z$ has no real solution, and therefore the sum in (5.1)
1077 is empty. Hence

$$1078 \quad (\mathcal{L}_T \rho)(z) = 0.$$

1079 This completes the proof. $\square$

1080 *Remark 5.3.* All results in this section are stated for $B > 0$, consistently with the
1081 reduction established in Remark 2.9. The case $B < 0$ follows immediately by applying
1082 the conjugacy $\psi(y) = -y$, which maps $T_{A,B}$ to $T_{-A,-B}$ and preserves all statistical
1083 properties of the invariant probability measure.

1084 The explicit form of the Perron-Frobenius operator in Theorem 5.2 has two imme-
1085 diate uses. First, it gives a closed functional equation for invariant densities, which is
1086 the starting point for both analytical and numerical approximation. Second, it makes
1087 the singularity mechanism completely transparent. The factor

$$1088 \quad |z - A|\sqrt{-\ln\left(\frac{z - A}{B}\right)}$$

vanishes at both ends of the image interval for different geometric reasons. As $z \rightarrow (A+B)^-$, the two inverse branches merge at the quadratic critical point, producing the fold singularity analyzed below. By contrast, as $z \rightarrow A^+$, the inverse images satisfy $|y_{\pm}(z)| \rightarrow \infty$. For densities supported on the invariant interval, their contributions vanish once these inverse images leave the support. The critical-value endpoint $A+B$ is therefore the relevant source of the post-critical square-root singularity.

Thus, the Perron-Frobenius operator formula is not merely formal; it encodes the local geometry of the map and the resulting statistical structure of its invariant probability measure, which makes it possible to identify the critical-value singularity generated by the quadratic fold. This is important for two reasons. First, it explains analytically the sharp peak observed in numerical histograms of long orbits (see Fig. 9 below). Second, it provides the first step toward a fuller description of the density as a superposition of singular contributions generated along the forward orbit of the critical value.

Theorem 5.2 gives the exact statistical evolution law for densities under the normalized Gaussian map (2.2). It is already enough to characterize invariant densities as fixed points of a linear operator equation in explicit closed form. The next statements record the most immediate consequences of this formula.

COROLLARY 5.4 (Invariant density equation). *Consider T in (2.2) with $B > 0$, and let ρ be an invariant density for T . Then ρ satisfies*

$$(5.8) \quad \rho(z) = \frac{\rho\left(\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right) + \rho\left(-\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right)}{2|z-A|\sqrt{-\ln\left(\frac{z-A}{B}\right)}},$$

for almost all $z \in (A, A+B)$, and $\rho(z) = 0$ for almost every $z \notin [A, A+B]$.

Proof. By invariance,

$$\rho = \mathcal{L}_T \rho.$$

Applying Theorem 5.2 gives exactly (5.8). Outside the image of T , the Perron-Frobenius operator vanishes, so any invariant density must also vanish there. $\square$

The invariant-density equation in (5.8) provides an explicit functional characterization of every invariant density of the normalized Gaussian map. Before analyzing its structural properties, it is instructive to examine a long-orbit empirical density obtained from a sufficiently long realization of the critical orbit.

Figure 9 provides a long-orbit histogram for $A = -1$ and $B = 2.25$, together with selected post-critical images $c_k = T^k(0)$. The alignment of several prominent peaks with these images is consistent with the singularity mechanism analyzed below. The figure is illustrative; the subsequent results establish the local singularity structure under the stated regularity assumptions.

COROLLARY 5.5 (Essential support of an invariant density). *Let ρ be an invariant density for T in (2.2) with $B > 0$. Then*

$$\text{ess supp } \rho \subset [A, A+B].$$

Proof. By Theorem 5.2,

$$(\mathcal{L}_T \rho)(z) = 0$$

for almost every $z \notin [A, A+B]$. Since ρ is invariant,

$$\rho = \mathcal{L}_T \rho \quad \text{in } L^1,$$

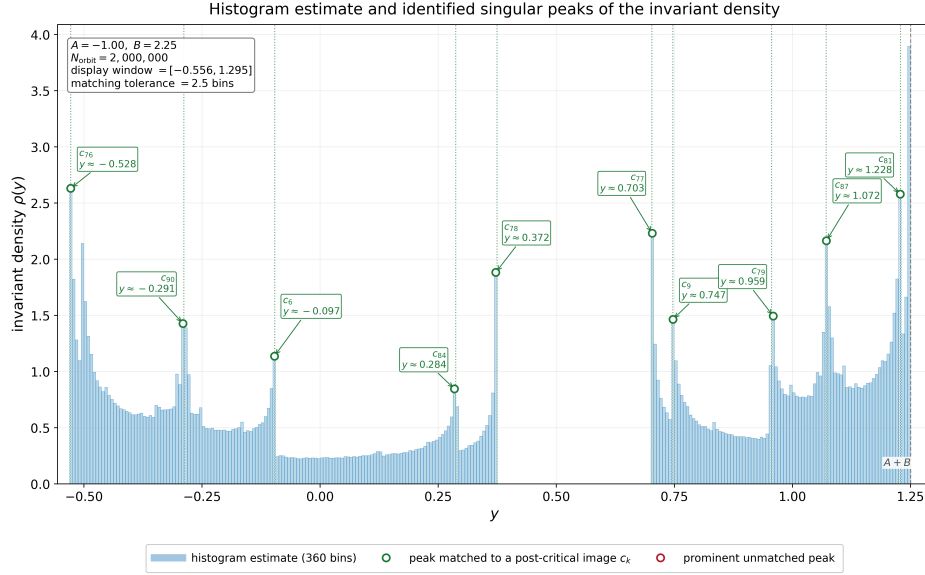


FIG. 9. Long-orbit empirical density of the normalized Gaussian map $T(y) = A + Be^{-y^2}$ for $A = -1$ and $B = 2.25$. The histogram is computed from a long post-transient critical orbit and approximates the long-orbit empirical density. The dominant peaks are automatically detected and associated with nearby post-critical images $c_k = T^k(0)$, shown by the vertical dotted lines. The correspondence is consistent with the post-critical singularity mechanism established analytically under the hypotheses of Theorem 5.9.

and therefore

$$\rho(z) = 0 \quad \text{for almost every } z \notin [A, A+B].$$

Hence, by definition of essential support,

$$\text{ess supp } \rho \subset [A, A+B]. \quad \square$$

COROLLARY 5.6 (Symmetry reduction). Consider T in (2.2) with $B > 0$, and let $\rho \in L^1(\mathbb{R})$. Define the even part of ρ by

$$\rho_e(y) = \frac{\rho(y) + \rho(-y)}{2}.$$

Then

$$(\mathcal{L}_T \rho)(z) = \frac{\rho_e\left(\sqrt{-\ln\left(\frac{z-A}{B}\right)}\right)}{|z-A|\sqrt{-\ln\left(\frac{z-A}{B}\right)}}, \quad (5.9)$$

for almost every $z \in (A, A+B)$.

Proof. By Theorem 5.2,

$$(\mathcal{L}_T \rho)(z) = \frac{\rho(u(z)) + \rho(-u(z))}{2|z-A|u(z)}$$

1143 for almost every $z \in (A, A + B)$, where

$$1144 \quad u(z) = \sqrt{-\ln\left(\frac{z - A}{B}\right)}.$$

1145 By definition of the even part,

$$1146 \quad \rho(u(z)) + \rho(-u(z)) = 2\rho_e(u(z)).$$

1147 Substitution into the preceding formula gives (5.9). $\square$

1148 For general nonuniformly expanding unimodal maps, invariant densities are often
 1149 studied through inducing schemes, towers, bounded-variation spaces, or spectral de-
 1150 compositions [22, 2, 3]. Such methods establish existence and statistical properties but
 1151 need not yield elementary pointwise formulas for the density near each post-critical
 1152 image. However, for the map (2.2), the explicit inverse branches make it possible to
 1153 analyze this structure directly. The results below derive the singularity at the critical
 1154 value, propagate it along the post-critical orbit, and compute its amplitudes in terms
 1155 of the cumulative derivative growth.

1156 For the normalized Gaussian family (2.2) with $B > 0$, the unique critical point
 1157 is $y_{cr} = 0$, and its image $T(0) = A + B$ is the first post-critical location at which the
 1158 quadratic fold generates a density singularity. The next theorem shows that, under
 1159 a mild regularity assumption at the critical point, the singularity is necessarily of
 1160 square-root type.

1161 **THEOREM 5.7** (Critical-value singularity of an invariant density). *Consider T*
 1162 *in (2.2) with $B > 0$, and let $I = [A, A + B]$. Assume that $A < 0 < A + B$, so*
 1163 *that the unique critical point $y_{cr} = 0$ belongs to $\text{int}(I)$. Let ρ be an invariant density*
 1164 *for the restriction $T : I \rightarrow I$, that is,*

$$1165 \quad \mathcal{L}_T \rho = \rho \quad \text{in } L^1(I).$$

1166 Assume that ρ admits a representative, still denoted by ρ , which satisfies the invariant-
 1167 density equation (5.8) pointwise in a left neighborhood of $A + B$, and is continuous at
 1168 the critical point $y_{cr} = 0$, with

$$1169 \quad 0 < \rho(0) < \infty.$$

1170 Then, as $z \rightarrow (A + B)^-$,

$$1171 \quad (5.10) \quad \rho(z) \sim \frac{\rho(0)}{\sqrt{B}} \frac{1}{\sqrt{A + B - z}}.$$

1172 Equivalently,

$$1173 \quad (5.11) \quad \lim_{z \rightarrow (A+B)^-} \sqrt{A + B - z} \rho(z) = \frac{\rho(0)}{\sqrt{B}}.$$

1174

1175 *Proof.* By hypothesis, the chosen representative of ρ satisfies (5.8) pointwise for
 1176 $z < A + B$ sufficiently close to $A + B$.

1177 Hence we study this expression as $z \rightarrow (A + B)^-$. Set

$$1178 \quad \varepsilon := A + B - z > 0.$$

1179 Then $\varepsilon \rightarrow 0^+$ as $z \rightarrow (A + B)^-$, and

$$1180 \quad z - A = B - \varepsilon = B \left(1 - \frac{\varepsilon}{B}\right).$$

1181 Therefore

$$1182 \quad \frac{z - A}{B} = 1 - \frac{\varepsilon}{B}.$$

1183 We first analyze the logarithmic term. Since

$$1184 \quad -\ln(1 - u) \sim u \quad (u \rightarrow 0^+),$$

1185 it follows that

$$1186 \quad (5.12) \quad -\ln\left(\frac{z - A}{B}\right) = -\ln\left(1 - \frac{\varepsilon}{B}\right) \sim \frac{\varepsilon}{B}.$$

1187 Hence

$$1188 \quad (5.13) \quad \sqrt{-\ln\left(\frac{z - A}{B}\right)} \sim \sqrt{\frac{\varepsilon}{B}}.$$

1189 Next, define

$$1190 \quad u(z) := \sqrt{-\ln\left(\frac{z - A}{B}\right)}.$$

1191 Then $u(z) \rightarrow y_{cr} = 0$ as $z \rightarrow (A + B)^-$. Since ρ is continuous at $y_{cr} = 0$,

$$1192 \quad \rho(u(z)) \rightarrow \rho(0), \quad \rho(-u(z)) \rightarrow \rho(0).$$

1193 Therefore the numerator in (5.8) satisfies

$$1194 \quad (5.14) \quad \rho(u(z)) + \rho(-u(z)) \rightarrow 2\rho(0).$$

1195 Now consider the denominator. Using $z - A = B - \varepsilon \rightarrow B$ together with (5.13),
1196 we obtain

$$1197 \quad (5.15) \quad 2(z - A)\sqrt{-\ln\left(\frac{z - A}{B}\right)} \sim 2B\sqrt{\frac{\varepsilon}{B}} = 2\sqrt{B}\sqrt{\varepsilon}.$$

1198 Substituting (5.14) and (5.15) into (5.8), we get

$$1199 \quad \rho(z) \sim \frac{2\rho(0)}{2\sqrt{B}\sqrt{\varepsilon}} = \frac{\rho(0)}{\sqrt{B}} \frac{1}{\sqrt{\varepsilon}}.$$

1200 Since $\varepsilon = A + B - z$, this yields

$$1201 \quad \rho(z) \sim \frac{\rho(0)}{\sqrt{B}} \frac{1}{\sqrt{A + B - z}},$$

1202 which is (5.10). Multiplying both sides by $\sqrt{A + B - z}$ gives (5.11). $\square$

Under the hypotheses of Theorem 5.7, the quadratic fold necessarily generates a one-sided square-root singularity at the critical value $c_1 = A + B$. This is the first singularity generated by forward propagation of the critical point. The statement is conditional on the existence of an invariant density that is finite, continuous, and strictly positive at $y = 0$.

A first consequence is that the singularity is integrable. Thus, the invariant density may diverge at the critical value while remaining normalizable.

COROLLARY 5.8 (Integrability of the critical-value singularity). *Under the hypotheses of Theorem 5.7, the singularity of ρ at $z = A + B$ is integrable.*

Proof. By Theorem 5.7,

$$\rho(z) \sim \frac{\rho(0)}{\sqrt{B}} \frac{1}{\sqrt{A+B-z}} \quad \text{as } z \rightarrow (A+B)^-.$$

Since the singularity of $(A+B-z)^{-1/2}$ is locally integrable at $z = A+B$, the same is true for ρ . Indeed, for every $\varepsilon > 0$,

$$\int_{A+B-\varepsilon}^{A+B} \frac{dz}{\sqrt{A+B-z}} = 2\sqrt{\varepsilon} < \infty.$$

□

Another useful consequence is qualitative. Theorem 5.7 identifies the critical value $A+B$ as the first singularity generated by the quadratic critical fold. This is the starting point for the propagation mechanism that generates further spikes along the forward orbit of $A+B$.

Theorem 5.7 establishes the one-sided square-root singularity generated at the critical value $c_1 = A+B$. Before deriving the general propagation law, it is instructive to visualize how repeated application of the Perron-Frobenius operator transforms an initially smooth density into one exhibiting the characteristic post-critical singularity cascade.

Figure 10 illustrates this mechanism by iterating the exact Perron-Frobenius operator associated with the normalized Gaussian map. Starting from a smooth probability density, the first iterate creates the singularity at the critical value, while subsequent iterates transport this singularity along the forward orbit of the critical point and generate new peaks at the post-critical images

$$c_k = T^k(0).$$

The figure therefore provides a direct numerical illustration of the dynamical mechanism that is formalized in the following theorem.

THEOREM 5.9 (Propagation of singularities along the critical orbit). *Consider T in (2.2) with $B > 0$, and define the forward orbit of the critical point by*

$$c_0 = 0, \quad c_{n+1} = T(c_n), \quad n \geq 0.$$

Let ρ be an invariant density for T satisfying the hypotheses of Theorem 5.7.

Fix $N \geq 1$, and assume that the post-critical points

$$c_1, \dots, c_N$$

are pairwise distinct and that

$$T'(c_n) \neq 0, \quad 1 \leq n \leq N-1.$$

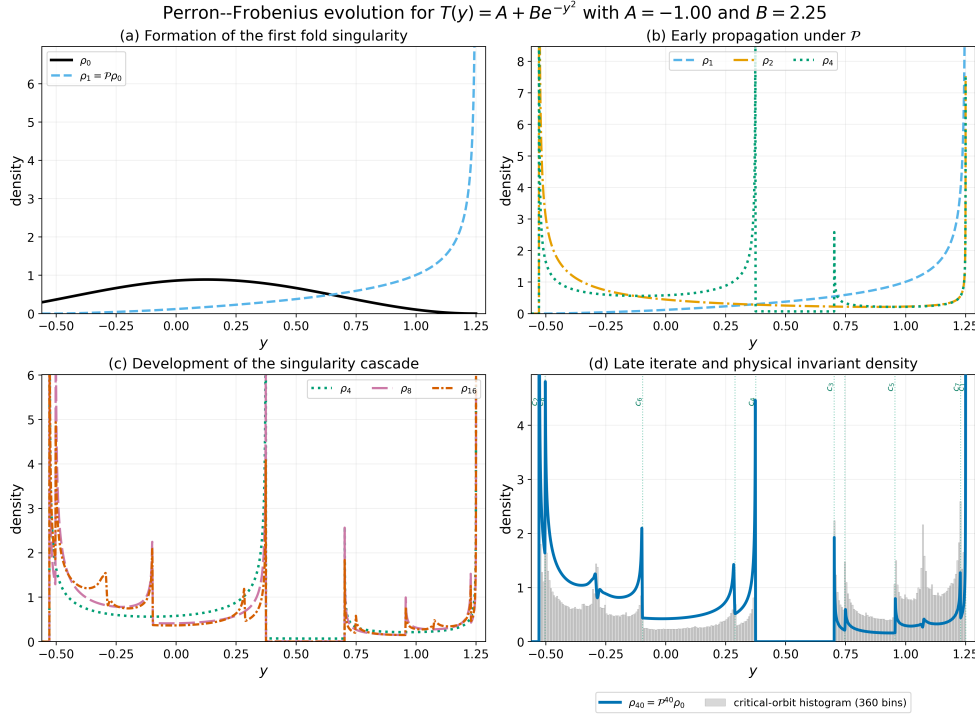


FIG. 10. Evolution of a smooth probability density under the Perron-Frobenius operator of the normalized Gaussian map $T(y) = A + Be^{-y^2}$ for $A = -1$ and $B = 2.25$. The operator is evaluated directly using the explicit inverse branches $x_{\pm}(y) = \pm\sqrt{-\log((y-A)/B)}$. (a) An initially smooth density ρ_0 develops a fold singularity after one application of the transfer operator. (b) The early iterates $\rho_n = \mathcal{L}_T^n \rho_0$ show the transport of this singularity through the phase interval. (c) Intermediate iterates exhibit the progressive development of a cascade of singular peaks. (d) A late transfer-operator iterate is shown together with an independently computed histogram of a long critical orbit. Vertical dotted lines indicate the first post-critical images $c_k = T^k(0)$. The figure illustrates how repeated application of the Perron-Frobenius operator generates singularities at successive images of the critical point.

Define

$$(5.16) \quad \sigma_1 = -1, \quad \sigma_{n+1} = \sigma_n \operatorname{sgn}(T'(c_n)), \quad 1 \leq n \leq N-1,$$

and

$$(5.17) \quad K_n(z) = \begin{cases} \frac{1}{\sqrt{\sigma_n(z - c_n)}}, & \sigma_n(z - c_n) > 0, \\ 0, & \sigma_n(z - c_n) < 0. \end{cases}$$

Assume additionally that the critical-value singularity admits a bounded remainder, namely,

$$(5.18) \quad \rho(z) = a_1 K_1(z) + \eta_1(z), \quad a_1 = \frac{\rho(0)}{\sqrt{B}} > 0,$$

in a punctured neighborhood of $c_1 = A + B$, where η_1 is bounded.

Assume also that, for each $1 \leq n \leq N - 1$, the contribution to the Perron-Frobenius operator from inverse branches other than the local inverse branch passing through c_n remains bounded in a neighborhood of c_{n+1} .

Then, for every $1 \leq n \leq N$, there exist a constant $a_n > 0$ and a function η_n , bounded in a neighborhood of c_n , such that

$$\rho(z) = a_n K_n(z) + \eta_n(z)$$

in a punctured neighborhood of c_n .

Moreover, the amplitudes satisfy

$$(5.18) \quad a_1 = \frac{\rho(0)}{\sqrt{B}}, \quad a_{n+1} = \frac{a_n}{\sqrt{|T'(c_n)|}}, \quad 1 \leq n \leq N - 1.$$

Consequently,

$$(5.19) \quad a_n = \frac{\rho(0)}{\sqrt{B}} \left(\prod_{k=1}^{n-1} |T'(c_k)| \right)^{-1/2}, \quad 2 \leq n \leq N.$$

Proof. By Theorem 5.7, the critical-value singularity occurs at

$$c_1 = T(c_0) = A + B$$

from the left. Hence its orientation is $\sigma_1 = -1$. By the additional bounded-remainder hypothesis,

$$\rho(z) = a_1 K_1(z) + \eta_1(z), \quad a_1 = \frac{\rho(0)}{\sqrt{B}},$$

where η_1 is bounded near c_1 .

Let $1 \leq n \leq N - 1$ and suppose that

$$\rho(y) = a_n K_n(y) + \eta_n(y) \quad \text{for all } y \in U \setminus \{c_n\}$$

with U being a neighborhood of c_n where η_n is bounded.

Since $T'(c_n) \neq 0$, the inverse function theorem implies that T is locally invertible near c_n . Thus there exists a neighborhood W of $c_{n+1} = T(c_n)$ with a unique inverse branch ϕ satisfying

$$\phi(c_{n+1}) = c_n, \quad T(\phi(z)) = z \quad \text{for all } z \in W \setminus \{c_{n+1}\}.$$

Applying the Perron-Frobenius formula (5.1) gives

$$(\mathcal{L}_T \rho)(z) = \frac{\rho(\phi(z))}{|T'(\phi(z))|} + R(z),$$

where $R(z)$ collects the contributions of all other preimages. By hypothesis, R remains bounded near c_{n+1} .

Substituting the local form of ρ yields

$$\frac{\rho(\phi(z))}{|T'(\phi(z))|} = \frac{a_n K_n(\phi(z))}{|T'(\phi(z))|} + \text{bounded terms.}$$

Using the Taylor expansion of T at c_n ,

$$T(y) = c_{n+1} + T'(c_n)(y - c_n) + O(|y - c_n|^2),$$

we obtain

$$\phi(z) - c_n = \frac{z - c_{n+1}}{T'(c_n)} + O(|z - c_{n+1}|^2).$$

Therefore, the singular side is transported according to

$$\sigma_{n+1} = \sigma_n \operatorname{sgn}(T'(c_n)).$$

Moreover,

$$\frac{K_n(\phi(z))}{|T'(\phi(z))|} = \frac{1}{\sqrt{|T'(c_n)|}} K_{n+1}(z) + \text{bounded terms}.$$

Because $\rho = \mathcal{L}_T \rho$, this gives

$$\rho(z) = a_{n+1} K_{n+1}(z) + \eta_{n+1}(z),$$

with

$$a_{n+1} = \frac{a_n}{\sqrt{|T'(c_n)|}},$$

and η_{n+1} bounded. Iterating this recurrence yields (5.19). $\square$

COROLLARY 5.10 (Explicit amplitudes for the Gaussian map). *Fix $N \geq 2$, and assume the hypotheses of Theorem 5.9 for this value of N . Then, for every $2 \leq n \leq N$,*

$$a_n = \frac{\rho(0)}{\sqrt{B}} \left(\prod_{k=1}^{n-1} 2B |c_k| e^{-c_k^2} \right)^{-1/2},$$

where

$$c_0 = 0, \quad c_{k+1} = A + B e^{-c_k^2}, \quad k \geq 0.$$

Equivalently,

$$a_n = \frac{\sqrt{2} \rho(0)}{(2B)^{\frac{n}{2}}} \frac{1}{\sqrt{\prod_{k=1}^{n-1} |c_k|}} \exp \left(\frac{1}{2} \sum_{k=1}^{n-1} c_k^2 \right), \quad 2 \leq n \leq N.$$

Proof. By Theorem 5.9,

$$a_n = \frac{\rho(0)}{\sqrt{B}} \left(\prod_{k=1}^{n-1} |T'(c_k)| \right)^{-1/2}, \quad 2 \leq n \leq N.$$

From (2.3), for each $k \geq 1$, we have

$$|T'(c_k)| = 2B |c_k| e^{-c_k^2}.$$

Substituting this into the product formula above gives

$$a_n = \frac{\rho(0)}{\sqrt{B}} \left(\prod_{k=1}^{n-1} 2B |c_k| e^{-c_k^2} \right)^{-1/2},$$

which is (5.20).

Since

$$\prod_{k=1}^{n-1} 2B |c_k| e^{-c_k^2} = (2B)^{n-1} \prod_{k=1}^{n-1} |c_k| e^{-c_k^2},$$

we obtain

$$a_n = \frac{\rho(0)}{\sqrt{B}} (2B)^{-\frac{n-1}{2}} \left(\prod_{k=1}^{n-1} |c_k| e^{-c_k^2} \right)^{-1/2},$$

that is,

$$a_n = \frac{\sqrt{2} \rho(0)}{(2B)^{\frac{n}{2}}} \left(\prod_{k=1}^{n-1} |c_k| e^{-c_k^2} \right)^{-1/2}.$$

Using the identity

$$\prod_{k=1}^{n-1} e^{-c_k^2} = \exp \left(- \sum_{k=1}^{n-1} c_k^2 \right),$$

the amplitude formula can be written as

$$a_n = \frac{\sqrt{2} \rho(0)}{(2B)^{\frac{n}{2}}} \frac{1}{\sqrt{\prod_{k=1}^{n-1} |c_k|}} \exp \left(\frac{1}{2} \sum_{k=1}^{n-1} c_k^2 \right).$$

This proves (5.21). $\square$

The explicit formula (5.21) shows that the amplitude of the singularities along the post-critical orbit is determined by two distinct mechanisms.

The first is the geometric prefactor

$$(2B)^{-n/2},$$

which depends only on the scale parameter B and decreases geometrically with n whenever $2B > 1$.

The second is the orbital factor

$$\frac{\exp \left(\frac{1}{2} \sum_{k=1}^{n-1} c_k^2 \right)}{\sqrt{\prod_{k=1}^{n-1} |c_k|}},$$

which reflects the geometry of the post-critical orbit. The exponential term grows whenever the orbit visits points with large $|c_k|$, while the product in the denominator amplifies the amplitudes whenever the orbit passes close to the origin, since small values of $|c_k|$ make the product $\prod |c_k|$ small and therefore increase its reciprocal.

Thus, the long-term behavior of the amplitudes is governed by the cumulative derivative along the post-critical orbit, as expressed by the recurrence

$$a_{n+1} = \frac{a_n}{\sqrt{|T'(c_n)|}}.$$

In particular, the decay or growth of the singularities is controlled by the dynamical properties of the orbit $\{c_n\}$ rather than by the parameter B alone.

The explicit amplitude formula derived in Corollary 5.10 provides a quantitative description of the singularity cascade generated along the post-critical orbit. In particular, the location of the n -th singularity is determined by

$$c_n = T^n(0),$$

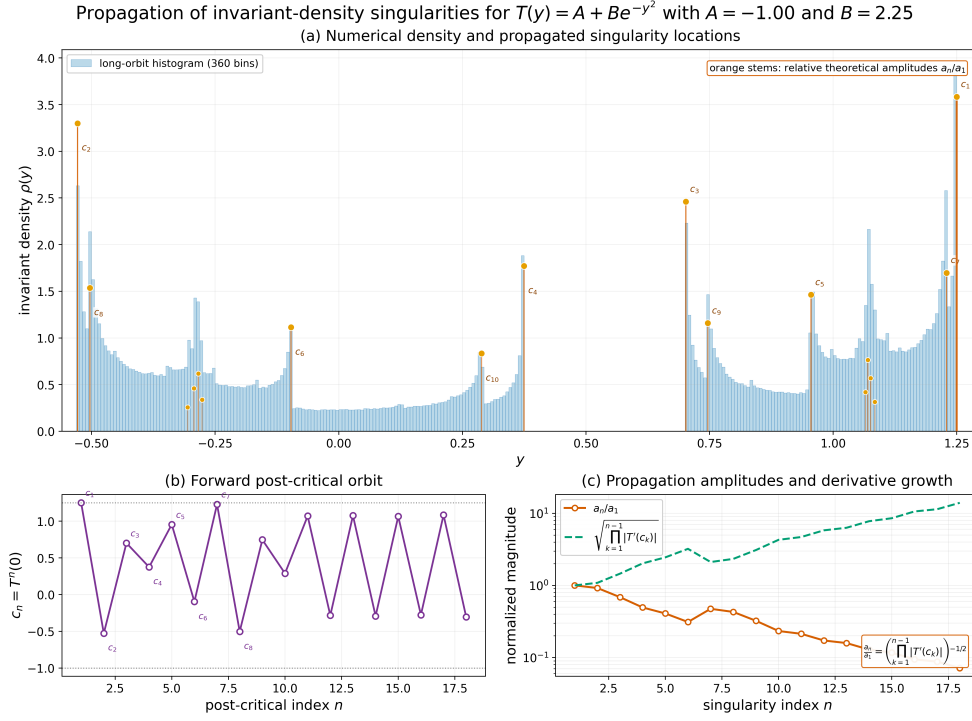


FIG. 11. *Post-critical organization of the empirical density and theoretical singularity amplitudes for the normalized Gaussian map $T(y) = A + Be^{-y^2}$, with $A = -1$ and $B = 2.25$. (a) Long-orbit empirical density together with selected post-critical locations $c_n = T^n(0)$, indicated by orange stems. The stem heights are proportional to the normalized theoretical amplitudes $\frac{a_n}{a_1} = \left(\prod_{k=1}^{n-1} |T'(c_k)|\right)^{-1/2}$. The alignment of prominent empirical peaks with post-critical locations is consistent with the singularity-propagation mechanism established under the hypotheses of Theorem 5.9. (b) Evolution of the post-critical sequence c_n , showing the successive locations visited by the critical orbit. (c) Normalized theoretical amplitudes a_n/a_1 and the corresponding square root of the cumulative derivative product $\sqrt{\prod_{k=1}^{n-1} |T'(c_k)|}$. Their reciprocal dependence illustrates the amplitude law of Theorem 5.9 and Corollary 5.10.*

1340 whereas its amplitude is controlled by the cumulative derivative growth experienced
1341 along the preceding segment of the critical orbit,

$$1342 \quad \frac{a_n}{a_1} = \left(\prod_{k=1}^{n-1} |T'(c_k)| \right)^{-1/2}.$$

1343 Figure 11 compares these theoretical predictions with a long-orbit empirical density.
1344 The figure separates the three components of the propagation mechanism: the
1345 observed empirical peaks, the evolution of the post-critical locations c_n , and the cor-
1346 responding amplitude modulation predicted by the derivative product.

1347 Theorems 5.7 and 5.9 show that the invariant density develops one-sided square-
1348 root singularities along the forward orbit of the critical point. The first theorem
1349 identifies the initial singularity at the critical value $c_1 = A + B$, while the second
1350 describes how such singularities propagate under the Perron-Frobenius operator along

the post-critical orbit. It is therefore natural to assemble these local asymptotic descriptions into a single representation. The next theorem makes this structure explicit: after subtracting the singular contributions generated by the post-critical orbit, the remaining part of the density is bounded.

THEOREM 5.11 (Finite post-critical decomposition). *Consider T in (2.2) with $B > 0$, and define the forward orbit of the critical point by*

$$c_0 = 0, \quad c_{n+1} = T(c_n), \quad n \geq 0.$$

Fix $N \geq 1$, and let ρ be an invariant density satisfying the hypotheses of Theorem 5.9 for this value of N .

Then there exist an open neighborhood $U \supset \{c_1, \dots, c_N\}$ and a bounded function $\psi_N : U \rightarrow \mathbb{R}$ such that

$$(5.22) \quad \rho(z) = \psi_N(z) + \sum_{n=1}^N a_n K_n(z), \quad z \in U \setminus \{c_1, \dots, c_N\},$$

where the amplitudes $a_n > 0$ are given by (5.18) and the orientations σ_n by (5.16).

Thus, near the finite set $\{c_1, \dots, c_N\}$, the singular contribution generated along the first N points of the post-critical orbit is a finite sum of one-sided square-root singularities, with amplitudes a_n and orientations σ_n .

Proof. By Theorem 5.9, for every $1 \leq n \leq N$ there exists a bounded function η_n such that

$$\rho(z) = a_n K_n(z) + \eta_n(z)$$

in a punctured neighborhood of c_n .

Since $c_1, \dots, c_N$ are pairwise distinct, we may choose bounded open intervals $U_1, \dots, U_N$ whose closures are pairwise disjoint, with $c_n \in U_n$, such that the above local representation holds on $U_n \setminus \{c_n\}$.

Set

$$U = \bigcup_{n=1}^N U_n$$

and define, for $z \in U \setminus \{c_1, \dots, c_N\}$,

$$\psi_N(z) = \rho(z) - \sum_{n=1}^N a_n K_n(z).$$

Fix $n \in \{1, \dots, N\}$. For $z \in U_n \setminus \{c_n\}$,

$$\psi_N(z) = (\rho(z) - a_n K_n(z)) - \sum_{\substack{m=1 \\ m \neq n}}^N a_m K_m(z).$$

The first term is bounded on U_n by the local representation supplied by Theorem 5.9. For $m \neq n$, the function K_m is bounded on U_n , because the neighborhoods are pairwise disjoint and hence $c_m \notin U_n$. Therefore ψ_N is bounded on $U_n \setminus \{c_n\}$.

Since there are only finitely many neighborhoods U_n , it follows that ψ_N is bounded on

$$U \setminus \{c_1, \dots, c_N\}.$$

Assigning arbitrary finite values to ψ_N at the points $c_1, \dots, c_N$ gives a bounded extension to all of U . Consequently,

$$\rho(z) = \psi_N(z) + \sum_{n=1}^N a_n K_n(z)$$

for every $z \in U \setminus \{c_1, \dots, c_N\}$, which proves (5.22). $\square$

The recurrence relation (5.18) shows that the evolution of the singularity amplitudes is governed by the cumulative derivative of the map along the post-critical orbit. This suggests introducing the Lyapunov exponent associated with the orbit of the critical value.

Consider the critical value orbit

$$c_0 = 0, \quad c_{n+1} = T(c_n).$$

The Lyapunov exponent of this orbit is defined by

$$\lambda_c := \lim_{n \rightarrow \infty} \frac{1}{n} \log |(T^n)'(c_1)| = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \log |T'(c_k)|,$$

whenever the limit exists. This quantity measures the average exponential rate of expansion of the map along the orbit of the critical value. As the next theorem shows, it also determines the asymptotic behavior of the singularity amplitudes generated along the post-critical orbit.

THEOREM 5.12 (Lyapunov control of the singularity amplitudes). *Assume that the hypotheses of Theorem 5.9 hold for every $N \geq 1$, so that the amplitudes a_n are defined for all $n \geq 1$. Assume furthermore that the finite Lyapunov exponent λ_c exists.*

Then the amplitudes $\{a_n\}$ satisfy

$$(5.23) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \log a_n = -\frac{\lambda_c}{2}.$$

Equivalently,

$$(5.24) \quad a_n = \exp\left(-\frac{\lambda_c}{2}n + o(n)\right).$$

In particular:

1. if $\lambda_c > 0$, then $a_n \rightarrow 0$ exponentially fast;
2. if $\lambda_c = 0$, then a_n has subexponential asymptotic size;
3. if $\lambda_c < 0$, then a_n grows exponentially.

Proof. By Theorem 5.9,

$$a_n = a_1 \left(\prod_{k=1}^{n-1} |T'(c_k)| \right)^{-1/2}.$$

Taking logarithms gives

$$\log a_n = \log a_1 - \frac{1}{2} \sum_{k=1}^{n-1} \log |T'(c_k)|.$$

1417 Dividing by n ,

$$1418 \quad \frac{1}{n} \log a_n = \frac{1}{n} \log a_1 - \frac{1}{2n} \sum_{k=1}^{n-1} \log |T'(c_k)|.$$

1419 As $n \rightarrow \infty$, the first term tends to 0, while the second tends to $-\lambda_c/2$.

1420 This proves (5.23).

1421 Equivalently,

$$1422 \quad \log a_n = -\frac{\lambda_c}{2} n + o(n),$$

1423 and exponentiating yields

$$1424 \quad a_n = \exp\left(-\frac{\lambda_c}{2} n + o(n)\right).$$

1425 This proves (5.24). The three alternatives follow immediately from the sign of λ_c . $\square$

1426 Theorem 5.12 shows that the long-run strength of the singularity cascade is controlled by the Lyapunov exponent of the post-critical orbit. Thus the geometric propagation law for the spikes is asymptotically equivalent to an exponential law governed by the average logarithmic derivative along the critical value orbit.

1430 COROLLARY 5.13 (Explicit form of the post-critical Lyapunov exponent). *For*
1431 *T in (2.2) with $B > 0$, let*

$$1432 \quad c_0 = 0, \quad c_{n+1} = T(c_n), \quad n \geq 0.$$

1433 *Under the hypotheses of Theorem 5.12, assume that the Lyapunov exponent of the*
1434 *post-critical orbit exists:*

$$1435 \quad \lambda_c = \lim_{n \rightarrow \infty} \frac{1}{n} \log |(T^n)'(c_1)|.$$

1436 *Then*

$$1437 \quad (5.25) \quad \lambda_c = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n (\log(2B) + \log |c_k| - c_k^2).$$

1438 *Equivalently,*

$$1439 \quad (5.26) \quad \lambda_c = \log(2B) + \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \log |c_k| - \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n c_k^2,$$

1440 *whenever the two latter limits exist.*

1441 *Proof.* By definition,

$$1442 \quad \lambda_c = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \log |T'(c_k)|.$$

1443 From (2.3) we obtain

$$1444 \quad |T'(c_k)| = 2B |c_k| e^{-c_k^2}.$$

1445 Taking logarithms gives

$$1446 \quad \log |T'(c_k)| = \log(2B) + \log |c_k| - c_k^2.$$

1447 Substituting this into the definition of λ_c yields (5.25). The split form (5.26) follows
1448 immediately whenever the two limits exist separately. $\square$

Corollary 5.13 shows that the post-critical Lyapunov exponent decomposes into three contributions: a constant term $\log(2B)$, a logarithmic term reflecting the proximity of the orbit to the critical point, and a quadratic term measuring its average excursion away from the origin. Thus the average expansion rate along the post-critical orbit is determined explicitly by the geometry of that orbit.

6. Discussion. The normalized Gaussian map shares the principal local features of smooth unimodal maps with a quadratic critical point: it is smooth, has a unique nondegenerate critical point, and has negative Schwarzian derivative away from that point. Its global geometry, however, is different from that of polynomial families. Since

$$T(y) \rightarrow A \quad \text{as} \quad |y| \rightarrow \infty,$$

the real line is mapped into the compact interval $[A, A + B]$ for $B > 0$. Thus, the nonlinear folding mechanism is localized near the critical point, while the global return is controlled by the exponentially decaying Gaussian profile. This distinction is reflected in the explicit parameter-space geometry derived in Sections 3 and 4.

A particularly useful feature of the Gaussian family is that its two monotone inverse branches are available in closed form. Consequently, the Perron-Frobenius operator can be written explicitly, and the connection between the quadratic critical fold and the singular structure of invariant densities becomes transparent. Under the hypotheses of Theorems 5.7 and 5.9, the critical value generates a one-sided square-root singularity whose location, orientation, and amplitude can be followed along finite post-critical segments. The amplitude law further relates the strength of these singularities to derivative growth along the critical orbit. These explicit formulas distinguish the Gaussian family from general smooth unimodal maps, for which analogous quantities are usually described through implicit inverse branches or more abstract functional-analytic constructions.

The analytical formulas also provide useful benchmarks for numerical computations. The explicit bifurcation curves can be compared directly with numerical parameter scans, the Collet-Eckmann family supplies examples with a prescribed post-critical itinerary and computable expansion rate, and the closed-form transfer operator eliminates numerical root finding when evolving densities. At the same time, the numerical density plots should be interpreted as illustrations of the analytical mechanisms rather than as pointwise convergence results, since narrow post-critical peaks are sensitive to discretization and histogram resolution.

6.1. Scope and limitations. The results obtained here do not constitute a complete classification of the dynamics of the Gaussian family. The analytical saddle-node and fixed-point period-doubling loci describe the fixed-point bifurcation structure, but the organization of higher-period windows, renormalization phenomena, and chaotic parameter sets remains open. Likewise, the explicit Collet-Eckmann construction identifies a concrete nonuniformly expanding family but does not determine the measure or complete geometry of the Collet-Eckmann parameter set.

The invariant-density results are also conditional on the regularity, noncollision, and bounded-remainder hypotheses stated in Section 5. A complete description of possible accumulations and interactions of post-critical singularities, as well as sharper global regularity and spectral properties of the associated transfer operator, would require additional analysis.

These limitations suggest natural extensions of the present work while preserving the main advantage of the Gaussian family: a combination of nontrivial unimodal dynamics with an unusually explicit analytical structure.

7. Conclusions. In this work, we have developed an analytical study of the generalized Gaussian iterated map through its reduction to the normalized family

$$T(y) = A + Be^{-y^2}.$$

The Gaussian structure yields explicit fixed-point saddle-node and period-doubling loci, a codimension-two cusp, the global contraction threshold, a complete fixed-point count, and a rigorous bound on the number of attracting periodic orbits.

We also constructed an explicit one-parameter family of maps satisfying the Collet-Eckmann condition and derived the Perron-Frobenius operator in closed form. Under the stated regularity and noncollision assumptions, the latter provides an explicit description of the one-sided square-root singularity generated at the critical value and its propagation along finite post-critical segments, with amplitudes controlled by derivative growth along the critical orbit.

A natural extension of the present work is the study of maps in which the nonlinear feedback term is given by a superposition of Gaussian functions, for example

$$(7.1) \quad f(x) = \beta + \sum_{j=1}^m \gamma_j e^{-\alpha_j (x-\delta_j)^2}, \quad \alpha_j > 0,$$

where $\beta, \alpha_j, \gamma_j, \delta_j \in \mathbb{R}$ are the model parameters.

Such systems arise in more complex models where multiple localized responses interact. In contrast with the single-Gaussian case, these maps are typically multimodal, with several critical points and a richer dynamical structure. The techniques developed here—based on the analysis of critical orbits, bifurcation curves, and invariant densities—provide a natural starting point for their investigation. In particular, understanding how interactions among multiple critical points influence the existence of Collet-Eckmann regimes, invariant measures, and singularity formation in densities remains an open and challenging problem.

This perspective suggests a broader program in which the analytical approach developed for the Gaussian map can be extended to a wider class of smooth nonlinear systems with multiple competing mechanisms.

Appendix A. Classical Results and Definitions. This section presents some classical results and definitions in dynamical systems that are essential for establishing the results derived in this paper. They are included here for completeness of theory presentation.

DEFINITION A.1 (Periodic orbit). *Let $T : I \rightarrow I$ be a map on an interval $I \subset \mathbb{R}$. A point $p \in I$ is a periodic point of period n if $T^n(p) = p$ and $T^k(p) \neq p$ for all $1 \leq k < n$, where T^n denotes the n -th iterate of T . The set*

$$\mathcal{O} = \{p, T(p), T^2(p), \dots, T^{n-1}(p)\}$$

is called a periodic orbit of period n . A periodic orbit of period 1 is called a fixed point.

DEFINITION A.2 (Basin of attraction). *Let $\mathcal{O}$ be an attracting periodic orbit of $T : I \rightarrow I$. The basin of attraction of $\mathcal{O}$ is the set*

$$\mathcal{B}(\mathcal{O}) = \left\{ y \in I : \text{dist}(T^n(y), \mathcal{O}) \rightarrow 0 \text{ as } n \rightarrow \infty \right\},$$

where $\text{dist}(y, \mathcal{O}) = \min_{p \in \mathcal{O}} |y - p|$. The basin $\mathcal{B}(\mathcal{O})$ is open in the relative topology of I , containing $\mathcal{O}$ itself.

DEFINITION A.3 (Immediate basin). The immediate basin $\mathcal{B}_0(\mathcal{O})$ is the union of the connected components of $\mathcal{B}(\mathcal{O})$ that contain points of $\mathcal{O}$.

DEFINITION A.4 (ω -limit set). Let $T : I \rightarrow I$ and $y_0 \in I$. The ω -limit set of y_0 under T is the set of all accumulation points of the forward orbit $\{T^n(y_0)\}_{n \geq 0}$, that is,

$$\omega(y_0) = \bigcap_{n \geq 0} \overline{\{T^k(y_0) : k \geq n\}}.$$

Equivalently, $p \in \omega(y_0)$ if and only if there exists a strictly increasing sequence of integers $n_k \rightarrow \infty$ such that $T^{n_k}(y_0) \rightarrow p$.

The ω -limit set captures the asymptotic behavior of the orbit of y_0 : it describes the set of points that the orbit visits infinitely often.

DEFINITION A.5 (Schwarzian derivative). Let $T : I \rightarrow \mathbb{R}$ be a C^3 map with $T'(y) \neq 0$. The Schwarzian derivative of T at y is

$$S(T)(y) = \frac{T'''(y)}{T'(y)} - \frac{3}{2} \left(\frac{T''(y)}{T'(y)} \right)^2.$$

THEOREM A.6 (Banach Fixed point Theorem). Let (X, d) be a complete metric space and let $f : X \rightarrow X$ be a contraction; that is, assume that there exists a constant $0 \leq L < 1$ such that

$$d(f(x), f(y)) \leq L d(x, y)$$

for all $x, y \in X$.

Then the following statements hold:

(i) The map f has a unique fixed point $x^* \in X$, that is,

$$f(x^*) = x^*.$$

(ii) For every initial point $x_0 \in X$, the sequence defined by

$$x_{n+1} = f(x_n), \quad n \geq 0,$$

converges to the fixed point x^* .

(iii) The convergence is at least geometric, in the sense that

$$d(x_n, x^*) \leq L^n d(x_0, x^*).$$

Proof. For the original proof, see [4]. A clear modern proof in the context of dynamical systems can be found, for example, in [18]. $\square$

THEOREM A.7 (Singer's theorem). Let $T : I \rightarrow I$ be a C^3 unimodal map on a compact interval I with negative Schwarzian derivative, then:

(i) the immediate basin of any attracting periodic orbit contains either a critical point of T or a boundary point of the interval I ;

(ii) each neutral periodic point is attracting;

(iii) there exists no interval of periodic points.

In particular, the number of non-repelling periodic orbits is bounded if the number of critical points of T is finite. Moreover, if all critical points of T are turning points, then T has at most two inessential periodic attractors (containing a boundary point of ∂I in its basin).

Proof. See ref. [14] for the proof of the present statement. The original formulation and proof are available in [36]. $\square$

THEOREM A.8 (Absolutely continuous invariant measure under Collet–Eckmann growth). *Let $T : I \rightarrow I$ be a C^3 unimodal map of a compact interval into itself, with a unique nonflat critical point $y_{\text{cr}} \in \text{int}(I)$ of order $\ell > 1$, and negative Schwarzian derivative on $I \setminus \{y_{\text{cr}}\}$. Assume that there exist constants $C > 0$ and $\Lambda > 1$ such that*

$$|(T^n)'(T(y_{\text{cr}}))| \geq C\Lambda^n, \quad n \geq 1.$$

Then T admits an ergodic absolutely continuous invariant probability measure μ .

Proof. The Collet–Eckmann condition implies

$$\sum_{n=1}^{\infty} |(T^n)'(T(y_{\text{cr}}))|^{-1/\ell} \leq C^{-1/\ell} \sum_{n=1}^{\infty} \Lambda^{-n/\ell} < \infty.$$

Thus the summability condition of Nowicki and van Strien [28] is satisfied. Their theorem therefore yields an absolutely continuous invariant probability measure, and the corresponding S -unimodal result of Nowicki and van Strien [27] gives ergodicity. $\square$

Code repository. Numerical simulations reported in this work were performed using the codes available at <https://americocunhajr.github.io/GaussMap>.

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REFERENCES

- [1] K. T. ALLIGOOD, T. D. SAUER, AND J. A. YORKE, *Chaos: An Introduction to Dynamical Systems*, Springer, 1996, <https://doi.org/10.1007/b97589>.
- [2] V. BALADI, *Positive Transfer Operators and Decay of Correlations*, World Scientific, 2000, <https://doi.org/10.1142/3657>.
- [3] V. BALADI AND D. SMANIA, *Linear response for smooth deformations of generic nonuniformly hyperbolic unimodal maps*, Ann. Sci. Éc. Norm. Supér., 45 (2012), pp. 861–926, <https://doi.org/10.24033/asens.2179>.
- [4] S. BANACH, *Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales*, Fundamenta Mathematicae, 3 (1922), pp. 133–181.
- [5] M. BENEDICKS AND M. VIANA, *Solution of the basin problem for Hénon-like attractors*, Invent. Math., 143 (2001), pp. 375–434, <https://doi.org/10.1007/s002220000109>.
- [6] M. BENEDICKS AND M. VIANA, *Random perturbations and statistical properties of Hénon-like maps*, Ann. Inst. H. Poincaré Anal. Non Linéaire, 23 (2006), pp. 713–752, <https://doi.org/10.1016/j.anihpc.2004.10.013>.
- [7] D. BORIN, M. ROLIM SALES, E. D. LEONEL, AND D. F. M. DE OLIVEIRA, *On the dynamics of the q -Tsallis Gauss iterated map*, Phys. A, 686 (2026), p. 131352, <https://doi.org/10.1016/j.physa.2026.131352>.
- [8] J. BOUVRIE AND B. HAMZI, *Kernel methods for the approximation of some key quantities of nonlinear systems*, J. Comput. Dyn., 4 (2017), pp. 1–19, <https://doi.org/10.3934/jcd.2017001>.

- [9] A. BOYARSKY AND P. GÓRA, *Laws of Chaos: Invariant Measures and Dynamical Systems in One Dimension*, Birkhäuser, 1997.
- [10] J. S. CÁNOVAS AND M. MUÑOZ-GUILLERMO, *On the dynamics of the q -deformed Gaussian map*, Int. J. Bifurc. Chaos, 30 (2020), p. 2030021, <https://doi.org/10.1142/S0218127420300219>.
- [11] B. V. CHIRIKOV, *A universal instability of many-dimensional oscillator systems*, Phys. Rep., 52 (1979), pp. 263–379, [https://doi.org/10.1016/0370-1573\(79\)90023-1](https://doi.org/10.1016/0370-1573(79)90023-1).
- [12] P. COLLET AND J.-P. ECKMANN, *Positive liapunov exponents and absolute continuity for maps of the interval*, Ergod. Theory Dyn. Syst., 3 (1983), pp. 13–46, <https://doi.org/10.1017/S0143385700001802>.
- [13] P. COLLET AND J.-P. ECKMANN, *Iterated Maps on the Interval as Dynamical Systems*, Birkhäuser, 2009, <https://doi.org/10.1007/978-0-8176-4927-2>.
- [14] W. DE MELO AND S. VAN STRIEN, *One-Dimensional Dynamics*, Springer-Verlag, Berlin, 1993.
- [15] J. A. DE OLIVEIRA, H. M. J. DE MENDONÇA, A. A. A. DA SILVA, AND E. D. LEONEL, *Critical slowing down at a fold and a period doubling bifurcations for a Gauss map*, Braz. J. Phys., 49 (2019), pp. 923–927, <https://doi.org/10.1007/s13538-019-00706-0>.
- [16] R. L. DEVANEY, *An Introduction to Chaotic Dynamical Systems*, Chapman and Hall/CRC, 3 ed., 2024.
- [17] M. HÉNON, *A two-dimensional mapping with a strange attractor*, Comm. Math. Phys., 50 (1976), pp. 69–77, <https://doi.org/10.1007/BF01608556>.
- [18] M. W. HIRSCH, S. SMALE, AND R. L. DEVANEY, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, Academic Press, 3 ed., 2013.
- [19] M. V. JAKOBSON, *Absolutely continuous invariant measures for one-parameter families of one-dimensional maps*, Comm. Math. Phys., 81 (1981), pp. 39–88, <https://doi.org/10.1007/BF01941800>.
- [20] G. KELLER, *Stochastic stability in some chaotic dynamical systems*, Monatsh. Math., 94 (1982), pp. 313–333, <https://doi.org/10.1007/BF01667385>.
- [21] G. KELLER AND C. LIVERANI, *Stability of the spectrum for transfer operators*, Ann. Sc. norm. super. Pisa, 28 (1999), pp. 141–152, https://www.numdam.org/item/ASNSP_1999_4_28_1_141_0.
- [22] G. KELLER AND T. NOWICKI, *Spectral theory, zeta functions and the distribution of periodic points for collet-eckmann maps*, Comm. Math. Phys., 149 (1992), pp. 31–69, <https://doi.org/10.1007/BF02096623>.
- [23] A. LASOTA AND J. A. YORKE, *On the existence of invariant measures for piecewise monotonic transformations*, Trans. Amer. Math. Soc., 186 (1973), pp. 481–488, <https://doi.org/10.2307/1996575>.
- [24] R. M. MAY, *Simple mathematical models with very complicated dynamics*, Nature, 261 (1976), pp. 459–467, <https://doi.org/10.1038/261459a0>.
- [25] D. NESIC AND A. TEEL, *A framework for stabilization of nonlinear sampled-data systems based on their approximate discrete-time models*, IEEE Trans. Automat. Control, 49 (2004), pp. 1103–1122, <https://doi.org/10.1109/TAC.2004.831175>.
- [26] T. NOWICKI AND S. VAN STRIEN, *Absolutely continuous invariant measures for c^2 unimodal maps satisfying the collet-eckmann conditions*, Invent. Math., 93 (1988), pp. 619–635, <https://doi.org/10.1007/BF01410202>.
- [27] T. NOWICKI AND S. VAN STRIEN, *Non-hyperbolicity and invariant measures for unimodal maps*, Ann. Inst. H. Poincaré Phys. Théor., 53 (1990), pp. 427–429.
- [28] T. NOWICKI AND S. VAN STRIEN, *Invariant measures exist under a summability condition for unimodal maps*, Invent. Math., 105 (1991), pp. 123–136.
- [29] E. OTT, C. GREBOGI, AND J. A. YORKE, *Controlling chaos*, Phys. Rev. Lett., 64 (1990), pp. 1196–1199, <https://doi.org/10.1103/PhysRevLett.64.1196>.
- [30] V. PATIDAR AND K. K. SUD, *A comparative study on the co-existing attractors in the Gaussian map and its q -deformed version*, Commun. Nonlinear Sci. Numer. Simul., 14 (2009), pp. 827–838, <https://doi.org/10.1016/j.cnsns.2007.10.015>.
- [31] F. PEREZ-CRUZ, S. VAN VAERENBERGH, J. J. MURILLO-FUENTES, M. LAZARO-GREDILLA, AND I. SANTAMARIA, *Gaussian processes for nonlinear signal processing: An overview of recent advances*, IEEE Signal Process. Mag., 30 (2013), pp. 40–50, <https://doi.org/10.1109/MSP.2013.2250352>.
- [32] G. PIANIGIANI, *Conditionally invariant measures and exponential decay*, J. Math. Anal. Appl., 82 (1981), pp. 75–88, [https://doi.org/10.1016/0022-247X\(81\)90226-2](https://doi.org/10.1016/0022-247X(81)90226-2).
- [33] W. M. RAHMAWATI AND F. LIANTONI, *Image compression and encryption using DCT and Gaussian map*, IOP Conf. Ser. Mater. Sci. Eng., 462 (2019), p. 012035, <https://doi.org/10.1088/1757-899X/462/1/012035>.
- [34] A. SAHAY AND C. PRADHAN, *Gauss iterated map based RGB image encryption approach*, in

- 1686 2017 International Conference on Communication and Signal Processing (ICCSP), 2017,
 1687 pp. 0015–0018, <https://doi.org/10.1109/ICCSP.2017.8286437>.
 1688 [35] SIMRAN AND V. V. M. S. CHANDRAMOULI, *Dynamics and chaos control of q -deformed Gauss-*
 1689 *ian map via superior approach*, *Chaos*, 36 (2026), p. 023108, [https://doi.org/10.1063/5.](https://doi.org/10.1063/5.0309958)
 1690 [0309958](https://doi.org/10.1063/5.0309958).
 1691 [36] D. SINGER, *Stable orbits and bifurcation of maps of the interval*, *SIAM J. Appl. Math.*, 35
 1692 (1978), pp. 260–267, <https://doi.org/10.1137/0135020>.
 1693 [37] S. H. STROGATZ, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology,*
 1694 *Chemistry, and Engineering*, Chapman and Hall/CRC, 3 ed., 2024.
 1695 [38] M. TOSIN, M. V. ISSA, D. MATOS, A. D. NASCIMENTO, AND A. CUNHA JR, *Employing 0-1 test*
 1696 *for chaos to characterize the chaotic dynamics of a generalized Gauss iterated map*, in Pro-
 1697 *ceedings of the XIV Conferência Brasileira de Dinâmica, Controle e Aplicações (DINCON*
 1698 *2019)*, 2019, <https://hal.science/hal-02388470>.
 1699 [39] M. VIANA, *Stochastic Dynamics of Deterministic Systems*, Instituto de Matemática Pura e
 1700 Aplicada (IMPA), Rio de Janeiro, 1997. Lecture Notes.
 1701 [40] J. I. YUZ AND G. C. GOODWIN, *Sampled-Data Models for Linear and Nonlinear Systems*,
 1702 Springer, 2014, <https://doi.org/10.1007/978-1-4471-5562-1>.